

8

Lithographic Control in Semiconductor Manufacturing

Historically, lithography engineering has focused on two key, complimentary aspects of lithographic quality: overlay performance and linewidth control. Linewidth (or critical dimension, CD) control generally means ensuring that the widths of certain critical features, measured at specific points on those features, fall within acceptable bounds. Overlay describes the positional errors in placing one mask layer pattern over an existing pattern on the wafer. Control of both CD and overlay is absolutely essential to producing high-yielding and high-performing semiconductor devices. And achieving adequate control becomes increasingly difficult as the feature sizes of the devices decrease.

8.1 Defining Lithographic Quality

How is lithographic quality defined? This simple question has a surprisingly complex answer. Lithography is such a large component of the total manufacturing cost of a chip and has such a large impact on final device performance that virtually all aspects of the lithography process must be carefully considered. Although somewhat arbitrary, the list below divides lithographic quality into four basic categories: photoresist profile control, overlay, downstream compatibility and manufacturability.

Photoresist Profile Control is a superset of the common CD control metric that is universally thought to be the most important aspect of high-resolution lithography processes. For many lithographic steps, the ability to print features at the correct dimensions has a direct and dramatic influence on device performance. It is typically measured as a mean to target CD difference for one or more specific device features, as well as a distribution metric such as the standard deviation. Spatial variations across the chip, field, wafer or lot are also important and can be characterized together or individually. In addition, the sensitivity of CD to process variations is often characterized and optimized as a method

to improve CD control. Metrics such as resolution, depth of focus and process latitude in general are expressions of CD control. For example, resolution can be defined as the smallest feature of a certain type that provides adequate CD control for a given process.

Profile control, however, recognizes that the printed resist patterns are three-dimensional in nature and a single CD value may not be sufficient to characterize their lithographic quality. Extension of CD control to profile control means taking into account other dimensions. In the 'z-direction' from the top to the bottom of the photoresist, the resist profile shape is usually characterized by a sidewall angle and a final resist height. In the 'x-y direction', patterns more complex than a line or space require a shape characterization that can include metrics such as corner rounding, line-end shortening, area fidelity, line edge roughness, or the critical shape difference (also called edge placement error).

Overlay is the ability to properly position one lithographic level with respect to a previously printed level. In one sense, lithography can be thought of as an effort to position photoresist edges properly on the wafer. But rather than characterize each edge individually, it is more convenient to correlate two neighboring edges and measure their distance from each other (CD) and the position of their midpoint (overlay). One simple reason for this division is that, for the most part, errors that affect CD do not influence overlay, and vice versa. (Unfortunately, this convenient assumption is becoming less and less true as the target feature sizes shrink.) Overlay is typically measured using special targets optimized for the task, but actual device structures can be used in some circumstances. Since errors in overlay are conveniently divided into errors influenced by the reticle and those influenced by the wafer and wafer stage, measurements are made within the exposure field and for different fields on the wafer to separate out these components. While historically CD control has gained the most attention as the limiter to feature size shrinks, overlay control is often just as critical.

Downstream Compatibility describes the appropriateness of the lithographic results for subsequent processing steps, in particular etch and ion implantation. Unlike many other processing steps in the manufacture of an integrated circuit, the handiwork of the lithographer rarely finds its way to the final customer. Instead, the true customers of the lithography process are the etch and implant groups, who then strip off those painstakingly prepared photoresist profiles when finished with them. Downstream compatibility is measured with such metrics as etch resistance, thermal stability, adhesion, chemical compatibility, strippability and pattern collapse.

Manufacturability is the final, and ultimate, metric of a lithographic process. The two major components of manufacturability are cost and defectivity. The importance of cost is obvious. What makes this metric so interesting, and difficult to optimize, is the relationship between cost and other metrics of lithographic quality such as CD control. While buying ultra flat wafers or upgrading to the newest stepper platform may provide an easy improvement in CD and overlay performance, the benefit may be negated by the cost increase. It is interesting to note that throughput (or more correctly overall equipment productivity) is one of the major components of lithographic cost for a fab that is at or near capacity due to the normal factory design that places lithography as the fab bottleneck.

Defectivity in all areas of the fab has been a major contributor to yield loss for most processes throughout the history of this industry. Because lithographic processes are

repeatedly applied to make a chip, any improvements in defectivity in the lithography area are multiplied many times over. Finally, concerns such as safety and environmental impact will always play a role in defining the overall manufacturability of a process.

The outline for defining lithographic quality presented above gives a flavor for the complexity of the task. Literally dozens of quality metrics are required to describe the true value of a lithographic result. However, once these metrics have been defined, their relative value to the customer evaluated, and methods for their measurement established, they become powerful tools for the continuous improvement required to remain competitive in semiconductor manufacturing.

8.2 Critical Dimension Control

The shrinking of the dimensions of an integrated circuit is the technological driver of Moore's Law (see Chapter 1). As the dimensions of a transistor shrink, the transistor becomes smaller, lighter, faster, consumes less power, and in many cases is more reliable. In addition, the semiconductor industry has historically been able to manufacture silicon devices at an essentially constant cost per area of processed silicon. Thus, an increased density of transistors per area of silicon results in huge cost advantages in the manufacture of chips. While many of these historical trends are being challenged (shrinking transistor size now often increases power consumption rather than reducing it), the compelling cost advantage of packing more transistors per unit area continues to push feature sizes smaller. However, as features become smaller, the need to control the quality of the lithographic results increases in proportion.

8.2.1 Impact of CD Control

Lithographers work hard to improve the control of the feature dimensions printed on the wafer for two very important reasons. First, and most obviously, the dimensions of device features affect the electrical behavior of those devices. In general, devices have been designed to work optimally over a very limited range of feature sizes. If those sizes become either too big or too small, a variety of undesirable consequences will result. Because of this, the very process of design itself must take into account the variability of the lithography process, resulting in the second important reason why linewidth control is important: its impact on the area of a designed chip.

Design rules are a complex set of geometric restrictions (such as the minimum allowed dimensions and spacings for each circuit element) that are used when creating a circuit layout (the design information that will eventually be translated into a photomask pattern). The overall goal of the design rules is to minimize the silicon area of the circuit while at the same time maintaining sufficient yield and circuit performance in the face of variations in the semiconductor manufacturing processes.

Consider a contact hole making contact to a source or drain of a transistor (Figure 8.1). One of the many design rules would define the minimum layout separation between the contact hole and the active area edge (dimension f in Figure 8.1). In general, this design rule would be determined by the contact to active layer overlay specification (O_{c-aa})

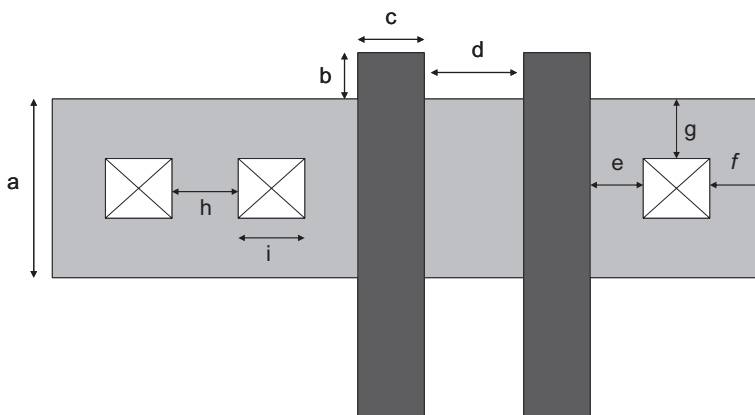


Figure 8.1 A simple layout showing a two-transistor structure with source/drain contact holes. One design rule would dictate the minimum allowed spacing between the edge of the contact and the edge of the active area (dimension f)

and the CD control specifications for the contact (ΔCD_c) and the active area width (ΔCD_{aa}).

$$\text{Minimum } f = O_{c-aa} + \Delta CD_c/2 + \Delta CD_{aa}/2 \quad (8.1)$$

Thus, the minimum spacing f can be reduced if the CD control specification is reduced. In general, though, the overlay specification is roughly three times the CD control specification so that overlay control represents about 75% of the control requirements and thus tends to dominate this design rule (see section 8.4). On the other hand, dimension h is completely determined by CD control considerations.

One of the classic examples of the influence of CD control is at the polysilicon gate level of standard CMOS logic devices. Physically, the polysilicon gate linewidth (paradoxically called the gate *length* by device engineers) controls the electrically important effective gate length (L_{eff}). In turn, L_{eff} is directly related to the transit time τ , the average time required to move an electron or hole from the source to the drain of the MOS device:

$$\tau = \frac{L_{\text{eff}}^2}{\mu V_{\text{ds}}} \quad (8.2)$$

where μ is the mobility of the charge carrier for the transistor (either holes or electrons), and V_{ds} is the drain–source voltage. The delay time of an unloaded inverter is about twice the transit time, plus any parasitic delays caused by resistance and capacitance of the device not associated with the gate. These parasitic delays tend to be about equal in magnitude to the delay caused by the gate transit time, but are independent of L_{eff} . Pass transistors have delays about equal to the inverter delay, and a two-input NAND gate has a delay about equal to twice that of an inverter. Given these timing constraints, a circuit's clock period has a minimum of about 10 times the delay given by Equation (8.2), but actual circuits use a slower 15 $\times$ multiplier to allow for variations in this delay time. Thus,

narrower gates tend to make transistors that can switch on and off at higher clock speeds. The squared dependence shown in Equation (8.2) means that small variations in the poly gate CD will cause a $2\times$ larger variation in device delay times.

Obviously, faster chips are more valuable than slower ones, as anyone who has priced a personal computer knows. But smaller is not always better. Transistors are designed (especially the doping levels and profiles) for a specific gate length. As the gate length gets smaller than this designed value, the transistor begins to ‘leak’ current when it should be off. If this leakage current becomes too high, the transistor is judged a failure. The ‘off’ current of a transistor, I_{off} , defined as the current when the gate source voltage $V_{\text{gs}} = 0$, is given by

$$I_{\text{off}} = I_{\text{ds}}(@ V_{\text{gs}} = V_{\text{th}}) e^{-V_{\text{th}}/s} \quad (8.3)$$

where I_{ds} is the drain–source current, V_{th} is the threshold voltage, and s is the temperature-dependent subthreshold swing (typically equal to about 80–90 mV/decade at room temperature, which is about 35–40 mV as used in the above equation). When $V_{\text{gs}} = V_{\text{th}}$, the device is not turned on and current is caused simply by diffusion. This diffusion current (I_{ds} in the above equation) will in turn be inversely proportional to L_{eff} . However, it is the exponential dependence of I_{off} on the threshold voltage that provides the strongest dependence of leakage current on gate length. Thus, ignoring the less important dependence of I_{ds} ,

$$\frac{d \ln I_{\text{off}}}{d V_{\text{th}}} \approx -1/s \quad (8.4)$$

The threshold voltage is dependent on the effective gate length due to what are called *short channel effects*. The change in threshold voltage from its long-channel value $V_{\text{th_long}}$ depends on the exact doping profiles used, but a typical simple model is

$$\Delta V_{\text{th}} = V_{\text{th_long}} - V_{\text{th}} = C_1 e^{-L_{\text{eff}}/2l} \quad (8.5)$$

where C_1 depends on the source-to-drain voltage and is on the order of 3–3.5 V, and l is called the *characteristic length* of the short channel effect and depends on the oxide thickness, the junction depth and the depletion layer width. This characteristic length is generally engineered to produce short-channel effects on the order of 0.1 V, so that $l \approx L_{\text{eff}}/7$. Thus,

$$\frac{d V_{\text{th}}}{d L_{\text{eff}}} = \Delta V_{\text{th}}/2l \approx 0.35 \text{ V}/L_{\text{eff}} \quad (8.6)$$

Combining Equations (8.6) and (8.4) using the chain rule,

$$\frac{d \ln I_{\text{off}}}{d \ln L_{\text{eff}}} \approx -0.35 \text{ V}/s \quad (8.7)$$

Assuming a typical subthreshold swing of 35 mV, a 10% reduction in L_{eff} will cause a 100% increase in I_{off} .

When printing a chip with millions of transistor gates, the gate lengths take on a distribution of values across the chip (Figure 8.2a). This across chip linewidth variation (ACLV) produces a range of transistor behaviors that affect the overall performance of

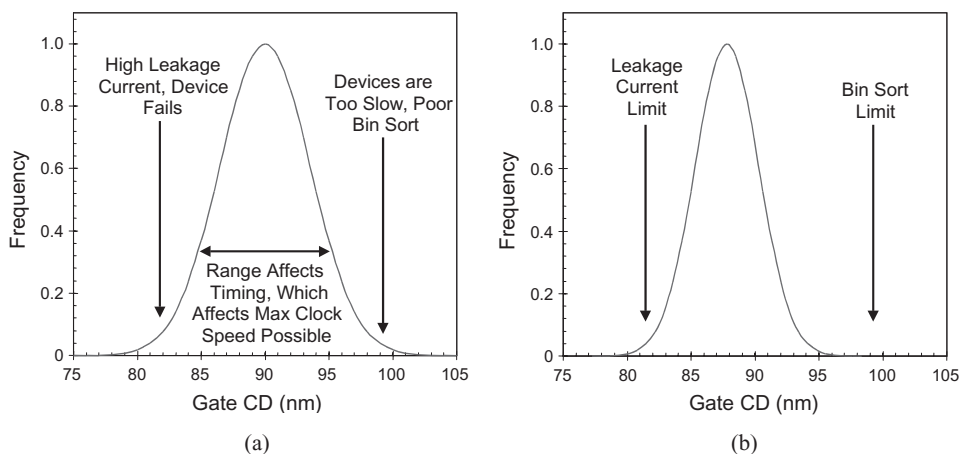


Figure 8.2 A distribution of polysilicon gate linewidths across a chip (a) can lead to different performance failures. Tightening up the distribution of polysilicon gate linewidths across a chip (b) allows for a smaller average CD and faster device performance

the chip. Although the specific details can be quite complicated and device specific, there are some very basic principles that apply.¹ As a signal propagates through the transistors of a chip to perform some operation, there will be several paths – connected chains of transistors – that operate in parallel and interconnect with each other. At each clock cycle, transistors are turned on and off, with the results passed to other interconnected transistors. The overall speed with which the operation can be performed (i.e. the fastest clock speed) is limited by the slowest (largest gate CD) transistors in the critical path for that operation (a critical path will typically have between 10 and 20 transistors in the path). Also, when gate lengths become too large, the drive current available to turn on or off the transistor may be too low, resulting in a potential reliability problem. On the other hand, the reliability of the chip is limited by the smallest gate CDs on the chip due to leakage current. If too many transistors on the chip have small gate CDs, the higher leakage current can result in unacceptably high power consumption.

The distribution of linewidths across the chip has another impact besides the limits of the largest and smallest CDs. The range of gate lengths produces a range of switching times for the transistors. If a group of connected transistors in a path all happens to be undersized (and thus faster), while another group of transistors in a path is oversized (and thus slower), a timing error can occur where the results from the two paths don't show up together at the same time. These timing errors, which are very difficult to predict and sometimes difficult to test for, result in device function errors. Today, interconnect delays (the time it takes a signal to travel from one transistor to another) are accounting for an ever larger share of the timing delays of critical paths for chips. Much design focus continues to be on evening out the interconnect delays among competing paths. Still, the gate delay distribution remains a critical part of the timing error problems on chips that are pushing the limits of clock speeds.

So how does improved CD control impact device performance? Obviously from the discussion above, a tighter distribution of polysilicon gate CDs will result in reduced

timing errors. This smaller range of linewidths also means that the average linewidth can be reduced without running into the leakage current limit (Figure 8.2b). As a result, the overall speed of the chip can be increased without impacting reliability. The resulting improved ‘bin sort’, the fraction of chips that can be put into the high clock speed bins upon final test of the device, can provide significant revenue improvements for the fab.

Like transistor gates, the impact of errors in the CD of a contact hole or via is different for oversized versus undersized contacts. If a contact hole is too big, the hole can overshoot the edge of the pattern below to which it is making contact, possibly causing an electrical short circuit. If a contact hole is too small, the electrical contact resistance will increase. Contact resistance is proportional to the area of the contact, and thus to the square of the contact diameter. A 10% decrease in contact CD causes a roughly 20% increase in contact resistance. If the contact/via resistance gets too high, signals propagating through that contact/via will slow down. For voltage-sensitive parts of the circuit (such as the source/drain contacts), the voltage drop across the contact can change the electrical characteristics of the device.

8.2.2 Improving CD Control

Fundamentally, errors in the final dimension of a feature are the result of errors in the tools, processes and materials that affect the final CD. An error in a process variable (the temperature of a hot plate, for example) propagates through to become an error in the final CD based on the various physical mechanisms by which the variable influences the lithographic result. In such a situation, a propagation of errors analysis can be used to help understand the effects. Suppose the influence of each input variable on the final CD were expressed in a mathematical form, such as

$$CD = f(v_1, v_2, v_3, \dots) \quad (8.8)$$

where v_i are the input (process) variables. Given an error in each process variable Δv_i , the resulting CD error can be computed from a high-order total derivative of the CD function in Equation (8.8).

$$\Delta CD = \sum_{n=1}^{\infty} \left(\Delta v_1 \frac{\partial}{\partial v_1} + \Delta v_2 \frac{\partial}{\partial v_2} + \dots \right)^n f(v_1, v_2, \dots) \quad (8.9)$$

This imposing summation of powers of derivatives can be simplified if the function is reasonably well behaved (and of course we hope that our critical features will be so) and the errors in the process variables are small (we hope this is true as well). In such a case, it may be possible to ignore the higher-order terms ($n > 1$), as well as the cross terms of Equation (8.9), to leave a simple, linear error equation (the first-order total derivative).

$$\Delta CD = \Delta v_1 \frac{\partial CD}{\partial v_1} + \Delta v_2 \frac{\partial CD}{\partial v_2} + \dots \quad (8.10)$$

Each Δv_i represents the magnitude of a *process error*. Each partial derivative $\partial CD / \partial v_i$ represents the *process response*, the response of CD to an incremental change in the variable. This process response can be expressed in many forms; for example, the inverse of the process response is called *process latitude*. Figure 8.3 shows a simple case where the

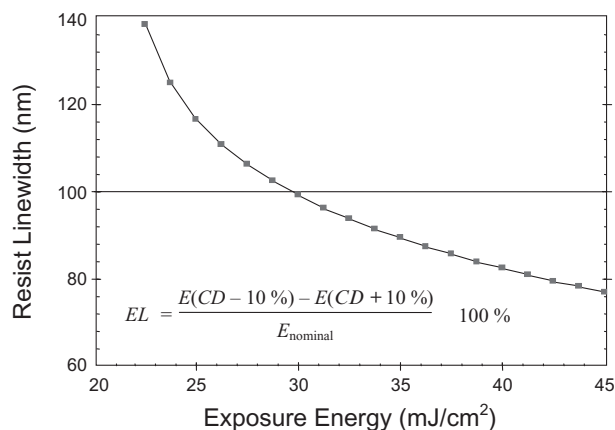


Figure 8.3 The common CD-versus-exposure dose (E) curve is used to measure exposure latitude (EL)

process variable is exposure dose and the common-use definition of exposure latitude (EL) is shown.

The linear error Equation (8.10) can be modified to account for the nature of the errors at hand. In general, CD errors are specified as a percentage of the nominal CD. For such a case, the goal is usually to minimize the relative CD error, $\Delta CD/CD$. Equation (8.10) can be put in this form as

$$\frac{\Delta CD}{CD} = \Delta v_1 \frac{\partial \ln CD}{\partial v_1} + \Delta v_2 \frac{\partial \ln CD}{\partial v_2} + \dots \quad (8.11)$$

Also, many sources of process errors result in errors that are a fraction of the nominal value of that variable (for example, illumination nonuniformity in a stepper produces a dose error that is a fixed percentage of the nominal dose). For such error types, it is best to modify Equation (8.11) to use a relative process error, $\Delta v_i/v_i$.

$$\frac{\Delta CD}{CD} = \frac{\Delta v_1}{v_1} \frac{\partial \ln CD}{\partial \ln v_1} + \frac{\Delta v_2}{v_2} \frac{\partial \ln CD}{\partial \ln v_2} + \dots \quad (8.12)$$

Although Equations (8.10)–(8.12) may seem obvious, even trivial, in their form, they reveal a very important truth about error propagation and the control of CD. There are two distinct ways to reduce ΔCD : reduce the magnitude of the individual process errors (Δv_i), or reduce the response of CD to that error ($\partial CD/\partial v_i$). The separation of CD errors into these two source components identifies the two important tasks that face the photolithography engineer. Reducing the magnitude of process errors is generally considered a *process control* activity and involves picking the right material and equipment for the job and ensuring that all equipment is working in proper order and all materials are meeting their specifications. Reducing the process response is a *process optimization* activity and involves picking the right process settings, as well as the right equipment and materials. Often, these two activities are reasonably independent of each other.

A note of caution: the derivation of Equation (8.10) assumed that the process errors were small enough to be linear and independent in their influence on CD. This will not always be the case in a real lithographic process. One need only consider the two variables of focus and exposure to see that the response of CD is certainly nonlinear and the two variables are highly dependent on each other (see section 8.5). Usually, a linear expansion such as that of Equation (8.10) is most useful as a guide to understanding rather than as a computational tool.

8.2.3 Sources of Focus and Dose Errors

The above sections describe why critical dimension (CD) control is important, and describe a common methodology for separating the causes of CD errors into two components: the magnitude of a process, material, or tool error, and the response of the lithographic process to that error. This separation of errors allows the process engineer to focus independently on the two methods for improving CD control: reducing the sources of errors and improving process latitude. In general, these approaches can be thought of as independent (though not always). Efforts to improve process latitude usually do not change the magnitude of the process errors, and vice versa. Let's consider the first approach, reducing the magnitude of the errors, using focus errors as an example.

Focus errors arise from many sources, both random and systematic. Tables 8.1 and 8.2 show estimates of the focus errors by source for different lithographic generations: a 1991 0.5- μm i-line process, 1995 0.35- μm i-line and KrF stepper processes, a 0.18- μm KrF scanner process, and a more modern 90-nm ArF process. The values shown are typical, but certainly vary from process to process. Although the tables are reasonably

Table 8.1 Examples of random focus errors (μm , 6σ) for different lithographic generations

Error Source	1991 ² i-line 0.50 μm	1995 ³ i-line 0.35 μm	1995 ³ KrF stepper 0.35 μm	2001 KrF scanner 0.18 μm	2005 ArF scanner 0.09 μm
Lens Heating (Compensated)	0.10	0.10	0.00	0.00	0.00
Environmental (Compensated)	0.20	0.20	0.10	0.10	0.05
Mask Tilt (actual/16)	0.05	0.05	0.10	0.05	0.05
Mask Flatness (actual/16)	0.12	0.12	0.12	0.12	0.07
Wafer Flatness (over one field)	0.30	0.33	0.33	0.15	0.07
Chuck Flatness (over one field)	0.14	0.03	0.03	0.03	0.03
Laser Bandwidth	0.0	0.0	0.20	0.1	0.04
Autofocus	0.20	0.08	0.10	0.07	0.04
Repeatability					
Best Focus	0.30	0.15	0.10	0.10	0.05
Determination					
Vibration	0.10	0.10	0.05	0.05	0.03
Total RSS Random Focus Errors	0.60	0.50	0.45	0.28	0.15

Table 8.2 Examples of systematic (μm , total range) and random focus error estimates combined to determine the Built-in Focus Errors (BIFE) of a process

Error Source	1991 ² i-line 0.50 μm	1995 ³ i-line 0.35 μm	1995 ³ KrF stepper 0.35 μm	2001 KrF scanner 0.18 μm	2005 ArF scanner 0.09 μm
Topography	0.5	0.3	0.3	0.10	0.05
Field Curvature and Astigmatism	0.4	0.4	0.3	0.08	0.05
Resist Thickness	0.2	0.2	0.2	0.10	0.05
Total Systematic Errors (range)	1.1	0.9	0.8	0.28	0.15
Total Random Errors (6σ)	0.60	0.50	0.45	0.28	0.15
Range/ σ	11	10.8	10.7	6	6
Total BIFE (6σ equivalent)	1.5	1.2	1.1	0.47	0.25

self-explanatory, a few items are worth noting. Errors in flatness or tilt on the mask are reduced by the reduction ratio of the imaging tool squared (assumed to be $4\times$ in this example). Wafer flatness is a total range over the exposure field assuming autoleveling is turned on. One of the primary advantages of the scanner is that the exposure slit sees a smaller region of the wafer and thus has a smaller focus error due to wafer nonflatness compared to a stepper, even for the same wafer. Note also that hot spots (focus errors caused by a particle on the backside of the wafer) and edge die problems (caused by errors in the autofocus system when attempting to focus at the edge of the wafer) are not included in this list.

Systematic focus errors include wafer topography (where chemical mechanical polishing has proven its worth, though is not yet perfect enough to make wafer topography a negligible problem), lens aberrations across the field or slit, and the errors due to focusing through the thickness of the resist. While not considered here, phase errors on a phase shift mask can also act like a focus error. Random and systematic errors can now be combined by assuming the systematic errors have a uniform probability distribution with the given range and convolving this uniform distribution with the Gaussian distribution of the random errors. Since in all cases the uniform width is large compared to the 3σ of the random errors, the result is a probability distribution that looks like the uniform distribution with error function tails. For an error function probability, the equivalent to a 6σ Gaussian range (99.7% probability) depends on the ratio of range/ σ . For range/ $\sigma = 11$, the total built-in focus errors (BIFE) of the process, with a probability of 99.7%, is range + 3.6σ . For range/ $\sigma = 6$, the BIFE is range + 4σ .

The exercise to fill in the values in tables such as those presented here serves two important functions. First, an evaluation of the BIFE can be combined with a measurement of the focus-exposure process window for the target process to see if the process capability (the process window) exceeds the process requirements (see section 8.5). Second, a listing of sources of focus errors inevitably leads to ideas where improvements can be made. The largest sources of errors (wafer nonflatness, for example, and wafer topography) offer the greatest potential for improvement.

Interestingly, a historical look at built-in focus errors reveals an interesting trend: the required depth of focus is about three times the minimum half-pitch for critical lithography layers such as gate polysilicon. While certainly not a rigorous analysis, using this rule of thumb can prove useful for projecting BIFE requirements to future generations of optical lithography.

Many errors that are not focus errors, but still affect CD, can be thought of as equivalent to an exposure dose error. Typical examples include:

- across-field (slit) intensity nonuniformity;
- field-to-field and wafer-to-wafer dose control;
- resist sensitivity variations (including development process variations);
- resist/BARC thickness variations (reflectivity variations);
- PEB time and temperature variations;
- flare variations;
- mask CD nonuniformity;
- errors in optical proximity correction (behaves like a mask CD error).

The required dose control has steadily tightened for smaller feature sizes. At the 0.5- μm generation, the built-in dose errors of a well-controlled process were about 15 % (range). At the 90-nm node, dose errors are typically less than about 6 %.

8.2.4 Defining Critical Dimension

A cross section of a photoresist profile has, in general, a very complicated two-dimensional shape (see Figure 8.4, for example). Measurement of such a feature to determine its width has many complications. Suppose, however, that we have been able to measure the shape of this profile exactly so that we have a complete mathematical description of its shape. How wide is it? The answer depends on how one *defines* the width. The original shape of the photoresist profile is simply too complex to be unambiguously characterized by a single width number. The definition of the width of a complex shape requires the definition of a *feature model*.⁴

A feature model is a mathematical function described by a conveniently small number of parameters used to appropriately depict the actual shape of the resist feature. For our application, one of these parameters should be related to the basic concept of the width of the resist profile. The most common feature model used for this application is a trapezoid (Figure 8.4). Thus, three numbers can be used to describe the profile: the width of

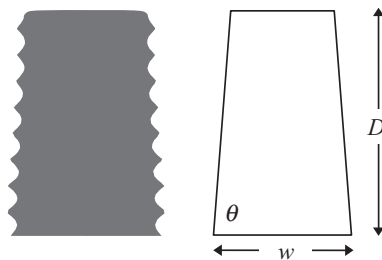


Figure 8.4 Example photoresist profile and its corresponding ‘best-fit’ trapezoidal feature model



Figure 8.5 Resist profiles at the extremes of focus show how the curvature of a pattern cross section can change

the base of the trapezoid (linewidth, w), its height (profile thickness, D) and the angle that the side makes with the base (sidewall angle, θ). To be perfectly general, the position of the feature (defined, for example, by the centroid of the feature model) can be specified and the shape can be made asymmetrical by allowing a different sidewall angle for each side.

Obviously, to describe such a complicated shape as a resist profile with just three numbers is a great simplification. One of the keys to success is to pick a method of fitting this feature model to the profile that preserves the important properties of the profile (and its subsequent use in the device). Thus, one can see that even given an exact knowledge of the actual photoresist profile, there are two potential sources of error in determining the critical dimension: the choice of the feature model and the method of fitting the feature model to the resist profile. Consider Figure 8.5, which shows resist profiles through focus exhibiting different curvatures of their sides (more on this topic in section 8.5). Using a trapezoidal feature model will obviously result in a less than perfect fit, which means that the criterion for best fit will influence the answer.

What is the best feature model and best method of fitting the feature model to measured data for a given application? Since the choice of the feature model is based both on relevance and convenience, and since the trapezoid is so commonly used for CD metrology, the impact of the feature model choice will not be discussed here. When fitting the feature model to the data, there are many possible methods. For example, one could find a best-fit straight line through the sidewall of the profile, possibly excluding data near the top and bottom of the profile. Alternately, one could force the trapezoid to always match the actual profile at some point of interest, for example at the bottom. Whenever the shape of the actual profile deviates significantly from the idealized feature model, the method of fitting can have a large impact on the results.

For example, as a lithographic process goes out of focus, the resist profile and the resulting feature size will change. But because the shape of the resist profile is deviating from a trapezoid quite substantially at the extremes of focus, CD can also be a strong function of how the profile data was fit. Figure 8.6 compares the measured CD through focus for two different feature model fitting schemes: a best-fit line through the sidewall and fitting the trapezoid to match the actual profile at a set threshold (height above the substrate). Near best focus, the two methods give essentially the same value since the resist profile is very close to a trapezoid. However, out of focus there can be a significant difference in the CD values ($> 5\%$) based only on the fitting method used.

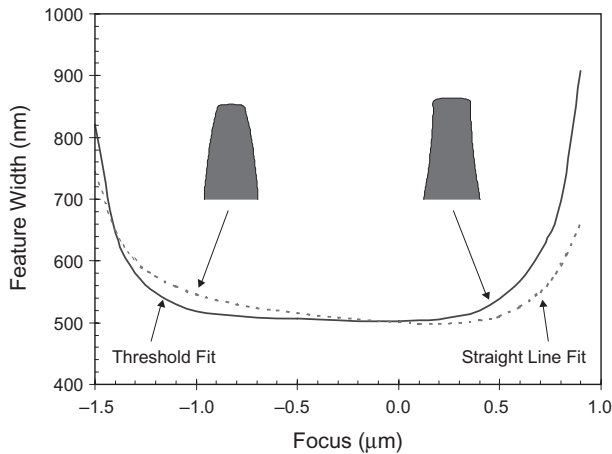


Figure 8.6 Using resist profiles at the extremes of focus as an example, the resulting measured feature size is a function of how the feature model is fit to the profile

In real metrology systems, the actual resist profile is never known. Instead, some signal (secondary electrons versus position, scattered light intensity versus wavelength) is measured that can be related to the shape of the resist profile. This signal is then fit to some model, which ultimately relates to the feature size being measured. Although a bit more complicated, the same principles still apply. Both the feature model and how that model is fit to the data will affect the accuracy of the results.

8.3 How to Characterize Critical Dimension Variations

Critical dimension (CD) errors in lithography come from many sources and exhibit many interesting characteristics. The classification of these errors into specific categories (systematic versus random, spatial versus temporal) is often the first step in a root cause analysis that allows the sources of CD variation to be identified and addressed. The nature of step-and-repeat or step-and-scan lithography on many identical wafers in a lot can produce many systematic spatial and temporal processing characteristics that impact CD errors. The discussion below will focus on step-and-scan lithography, with emphasis on the characterization of systematic error sources.

8.3.1 Spatial Variations

The systematic spatial variations of CD within a wafer exhibit four basic signatures: across-wafer, across-field in the scan direction, across-field in the slit direction, and across-field independent of the scan/slit orientation (Figure 8.7). The physical causes of variations in each of these spatial domains can be very different. Thus, dividing up the total spatial variation of CDs among these four components is the first step in identifying causes and remedies for unwanted CD errors.

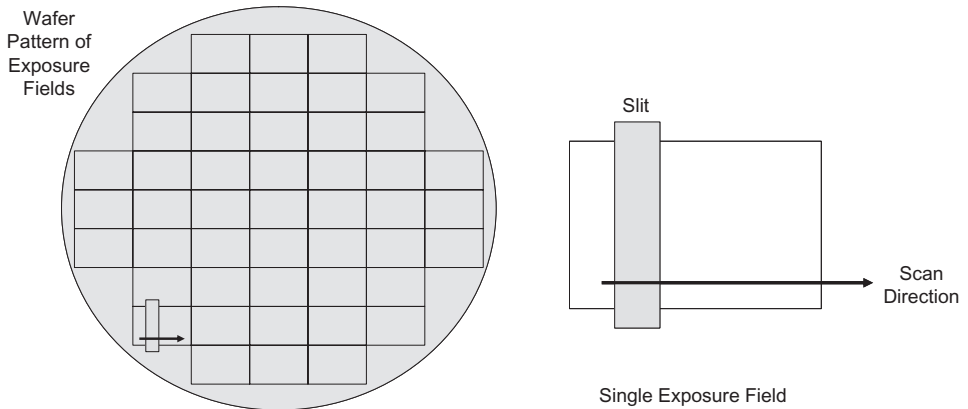


Figure 8.7 A wafer is made up of many exposure fields, each with one or more die. The field is exposed by scanning a slit across the exposure field

For those processing steps that result in a full-wafer treatment (resist coating, post-exposure baking, development), errors in that processing can contribute to CD variations with a full-wafer signature. Essentially all wafer processing other than exposure is full-wafer processing and results in an across-the-wafer variation in CD. Examples include:

- underlying filmstack thickness variations due to deposition;
- resist and bottom antireflection coating thickness uniformity;
- prebake temperature uniformity;
- post-exposure bake temperature uniformity;
- development uniformity (due to application, wetting, surface tension, etc.);
- first to last exposure shot variation due to lens heating or post-exposure delay;
- edge shot focus variation due to autofocus/autolevel problems at the edge of the wafer.

Many of these sources of error will exhibit mostly radial signatures, though other signatures (such as a wedge variation – a linear variation from one side of the wafer to the other) are certainly possible. The variation in CD caused by the difference in the first and last exposure shot can have a very different signature and, of course, depends on the specific stepping pattern used. Hot spots, an extended region of the wafer with higher or lower CDs, can be caused by backside defects or wafer chucking errors that produce a localized focus error that cannot be compensated for by autofocus and autoleveling mechanisms.

The process of scanning a slit in one direction across the field results in a distinct difference in the way many across-field error sources are manifest. Spatial variations across the field in the scan direction will average out to produce a fairly uniform signature in that direction. Spatial variations in the slit direction will be systematically reproduced everywhere in the field. There is one major systematic across-field CD variation source that is independent of scan orientation: reticle CD errors. Coupled with the mask error

enhancement factor (MEEF), reticle errors will systematically reproduce their signature across the wafer field without any averaging or other scan-induced changes.

Along the long dimension of the slit, there is no averaging effect of scanning to reduce the sources of CD variation. Thus, variations in the slit direction create a unique signature for the exposure tool. Common sources of CD errors with systematic variations in the slit direction include:

- lens aberrations;
- illumination (dose) uniformity;
- source shape (partial coherence) variations;
- flare.

In general, systematic errors in the slit direction dominate the systematic across-field errors.

Most of the systematic sources of CD errors that plague lithographers in the slit direction tend to average themselves out in the scan direction – this is one of the major benefits of step-and-scan technology. And while on-the-fly focusing can greatly reduce the impact of long-range wafer nonflatness, it can also induce some systematic CD errors in the scan direction. Autofocus systems can be fooled by underlying topography on the wafer (for example, when scanning past a logic region onto a dense memory region) and systematically but incorrectly adjust focus as it scans.

Variations from wafer to wafer can take on a spatial component due to systematic differences in how one wafer is processed versus another (it is important to differentiate the spatial variation being considered here from a temporal variation caused by a change in the processing of each wafer over time, to be discussed below). The main source of such variations comes from multiple processing chambers (modules) used per lot. For example, a single photolithography cell may have two developer cups and two post-exposure bake (PEB) hot plates. Any systematic difference between these track modules will give a systematic wafer-to-wafer error grouped by which module was used in the processing. Likewise, incoming wafers may have systematic wafer-to-wafer filmstack variations due to multiple deposition chambers.

Lot-to-lot systematic spatial variations are typically caused by differences in processing tools:

- different photolithography cells;
- different reticles;
- different deposition tools for incoming wafers;
- different metrology tools (tool matching errors).

Again, one should distinguish between those lot-to-lot differences caused by using different tools versus differences caused by temporal drifts in the behavior of a given tool.

8.3.2 Temporal Variations and Random Variations

Temporal variations can be seen both wafer-to-wafer and lot-to-lot, depending only on the timescales involved. Some effects, however, such as lens heating, have different within-lot and lot-to-lot signatures. Temporal variations can be either drifts in lithographic

inputs (a gradual barometric pressure change causing a focus shift) or an abrupt disturbance (switching to a new batch of photoresist). Although any lithographic input can conceivably have a temporal variation, some of the most common are:

- focus drift;
- dose calibration (intensity measurement) drift;
- resist batch change;
- laser center wavelength and bandwidth variations;
- lens heating;
- metrology drift;
- long-term lens degradation.

Of course, almost any lithographic input can have random variations associated with it. An important distinction between random errors and temporal variations as discussed above is the time constant of change. If a variable changes randomly but over a period of time such that many wafers can be processed (that is, between hours and months), such errors are generally classed as temporal variations. If, however, the variations are rapid enough to change across a wafer or from wafer to wafer, the errors are described as random. Some of the most significant random errors are:

- focus;
- dose;
- scan synchronization (x , y and z);
- PEB temperature/heating transients;
- metrology.

Of course, every input to the lithography process has some level of random variation associated with it.

8.3.3 Characterizing and Separating Sources of CD Variations

Different methods are often employed to characterize and control errors from different sources. For example, temporal variations are usually characterized using trend chart and statistical process control (SPC) techniques.⁵ Wafer-to-wafer and lot-to-lot variations caused by differences in processing tools or chambers can be caught using statistical correlation techniques (for example, where the average CD value obtained in one photolithography cell is compared to another and a statistically significant difference is flagged). The separation of CD errors into random and within-wafer spatially systematic is one of the most challenging problems in lithography data analysis. Further separation of the within-wafer systematic errors into their specific components (across-wafer, across-field, slit direction and scan direction) is both difficult and important. There are two basic approaches that can be used: characterizing output CD variations and characterizing input parameter variations.

The basic statistical technique for separating various systematic spatial variations and random errors is the use of the 'composite' spatial source. For example, the variation of CD across a wafer can have both a systematic component and a random component. By measuring a number of wafers at the exact same spatial locations, averaging of the measurements at each spatial location creates a composite wafer with the random errors

mostly averaged out. This composite wafer can then be used to characterize the systematic spatial signature of the error. Subtracting the composite wafer from the actual wafer data produces a residual set of random errors that can be characterized by a mean and standard deviation.

Let $M_i(x,y)$ be the measured CD on wafer i at the location (x,y) on the wafer for a measured group of N wafers. This measured value can be separated into systematic and random errors:

$$M_i(x,y) = CD_0 + S_i(x,y) + R_i(x,y) \quad (8.13)$$

where CD_0 is the nominal (target) CD, S_i is the systematic error for wafer i and R_i is the random error component. If the experiment used to make the measurements is conducted appropriately, the systematic wafer-to-wafer variations can be kept to a minimum (by making sure that each wafer is processed identically and by processing the wafers over a very short period of time). If this wafer-to-wafer variation is much smaller than the across-wafer variation, it is possible to approximate $S_i(x,y) \approx S(x,y)$ = the systematic across-wafer spatial signature. A 'composite' wafer CD error $CCDE(x,y)$ is created by averaging the measurements from the different wafers at each point:

$$CCDE(x,y) = \frac{1}{N} \sum_{i=1}^N M_i(x,y) - CD_0 \approx S(x,y) + \frac{1}{N} \sum_{i=1}^N R_i(x,y) \quad (8.14)$$

If normally distributed, the random errors will have a mean of zero at each point on the wafer. For a large enough sample N , the average of the sample will approximate the mean of the population with high accuracy. Thus,

$$\frac{1}{N} \sum_{i=1}^N R_i(x,y) \approx 0 \quad \text{with an uncertainty of } \sigma_R(x,y)/\sqrt{N} \quad (8.15)$$

where $\sigma_R(x,y)$ is the standard deviation of the random CD error at the point (x,y) . By approximating the standard deviation of the random error with the standard deviation of the residuals of the N sample of measurements $s_R(x,y)$, the accuracy of Equation (8.15) can be estimated. By picking N to be sufficiently large, one can use the composite wafer error as an estimate of the systematic across-wafer CD error signature within a known level of error.

$$S(x,y) \approx CCDE(x,y) \quad \text{with an estimated uncertainty of } s_R(x,y)/\sqrt{N} \quad (8.16)$$

In turn, the composite wafer can be analyzed to create a composite field. Averaging the same field measurement points from all fields across the wafer produces a signature that is characteristic of the field error only (with the assumption that the across-wafer errors vary slowly enough across the field as to contribute only negligibly to the composite field). Once the composite field is obtained, it can be subtracted from the composite wafer to create a separate across-wafer-only signature.

Likewise, the composite field can be broken down into a composite slit and composite scan signature. This separation into components is complicated by the fact that the contribution of reticle errors to the composite field is independent of scan direction. For one simplifying case, however, all components can be properly separated. If the reticle CD variations do not exhibit any systematic scan or slit direction components, subtraction of

the composite slit and composite scan from the composite field should yield the reticle contribution. It seems likely, however, that this simplifying assumption will often prove false. Any reticle error signature that has a systematic spatial signature (the edges of the reticle might have a lower CD than the middle, for example) will confound the separation of errors. In such a case, the second method of analyzing CD variations – a direct characterization of the sources of variation – can make this spatial decomposition even more meaningful.

While the previous paragraphs described a detailed statistical approach to the characterization of measured CD errors, a second approach would directly look for and measure the variations in lithographic inputs thought to cause these errors. Such a ‘sources of variations’ study can be coupled with the sensitivities of CD to each error source to predict the resulting CD error distributions. One of the most common examples is the measurement of reticle CD errors to create a spatial map of the reticle. By combining this reticle map with measured or simulated MEEF values for the specific features (mask error enhancement factor, see section 8.7), a predicted wafer CD error map due only to reticle errors can be generated.

The most effective means of characterizing CD errors in a lithographic process is a combination of a sources of variation study and statistical decomposition of measured wafer errors. Using the reticle CD error map described above, for example, the reticle contribution to the across-field CD errors from the scan and slit direction signatures can be subtracted out, making the statistical analysis of spatial variation more accurate. Also, by comparing a ‘bottoms up’ prediction of wafer CD errors through a sources of variations study with actual measured CD variations, one can check assumptions about which errors are most significant and which fab investments meant to improve CD control are most likely to pay off.

8.4 Overlay Control

Like critical dimension control, overlay control is a vital part of lithography in semiconductor manufacturing. Errors in the overlay of different lithographic levels can directly cause a number of electrical problems. At the gate level, overlay errors can pull a gate line end close to the active area, causing an increase in leakage current. At contact or via, overlay errors can reduce the overlap between the contact hole and the underlying conductor, causing higher contact resistance or, if bad enough, a short to the wrong electrical connection. On the design side, design features are spread out to be insensitive to expected errors in overlay. If overlay control can be improved, designs can be shrunk, allowing smaller die, more die per wafer and lower cost per die. As a result, economics dictates that overlay specifications must shrink in lockstep with device geometries, requiring continuous improvement in overlay measurement and control.

Overlay is defined as the positional accuracy with which a new lithographic pattern has been printed on top of an existing pattern on the wafer, measured at any point on the wafer. This is opposed to the slightly different concept of *registration*, which is the positional accuracy with which a lithographic pattern has been printed as compared to an absolute coordinate grid. While registration can be important (for example, when considering the positional accuracy of features on a photomask), overlay is both much easier to

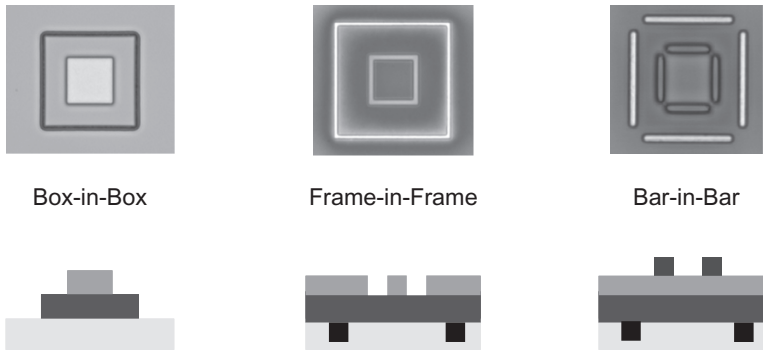


Figure 8.8 Typical 'box-in-box' style overlay measurement targets, showing top-down optical images along the top and typical cross-section diagrams along the bottom. The outer box is typically 20 μm wide. (Courtesy of KLA-Tencor Corp.)

measure and more relevant to lithographic quality and yield. In this section we'll examine some basics of how overlay is measured, and how overlay data are used.

8.4.1 Measuring and Expressing Overlay

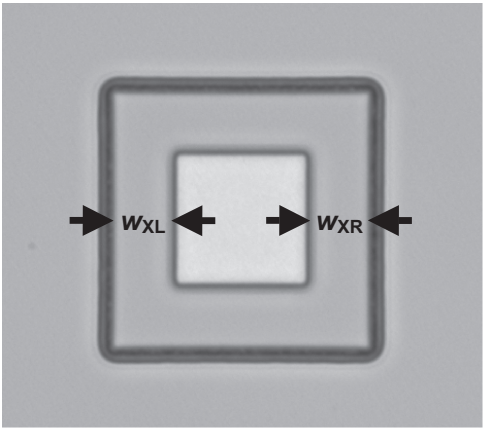
Overlay measurement involves the design of special patterns used on two different lithographic printing steps such that a metrology tool, when viewing the results of both printed patterns simultaneously, can measure errors in overlay for that point on the wafer. The most common pattern used today is the so-called box-in-box target (Figure 8.8). An outer box is printed during the first lithographic step and an inner box is printed during the second lithography pass.

Overlay measurement looks for differences in the space between the printed boxes (the distances between inner and outer box edges). For example, the x -overlay error would be one-half of the difference between the right and left widths between the inner and outer boxes (Figure 8.9). By using both left and right widths, processing errors that affect the dimensions of the boxes symmetrically will cancel out. Repeating this measurement for the top and bottom of the boxes gives the y -overlay error. The x - and y -overlay measurement pair produces a vector of the overlay error at the wafer and field location of the measurement.

While symmetrical errors in CD measurement will cancel when calculating overlay, asymmetrical errors will not. Coma in the overlay measurement optics, for example, can cause a left-right asymmetry in the measured box-in-box CDs, thus giving an error in the overlay measurement. Such an asymmetry can easily be detected by rotating the wafer by 180° and remeasuring the target. In the absence of any measurement asymmetries, the two measurements would be the same, but of opposite signs. Thus, the *tool-induced shift* (TIS) is defined as

$$TIS = \text{Overlay}(0^\circ) + \text{Overlay}(180^\circ) \quad (8.17)$$

with $TIS = 0$ for an ideal measurement. If TIS for an overlay measurement tool is fixed, it can be easily measured and subsequent overlay measurement corrected for this



$$x\text{-overlay} = 0.5(w_{XL} - w_{XR})$$

Figure 8.9 Measuring overlay as a difference in width measurements

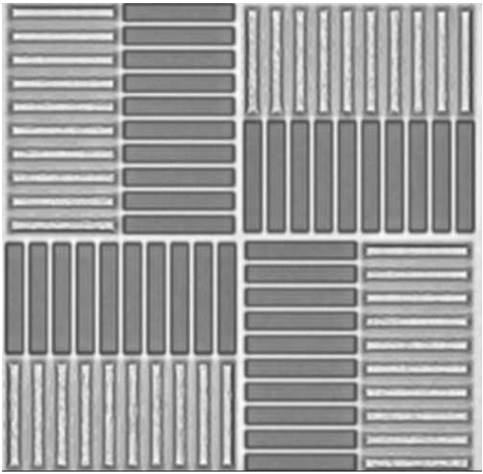


Figure 8.10 Typical AIM target where the inner bars (darker patterns in this photograph) are printed in one lithographic level and the outer (brighter) bars in another level. (Courtesy of KLA-Tencor Corp.)

systematic error. Thus, it is the variation of TIS that causes increased measurement uncertainty.

Asymmetries in the measurement target on the wafer can also cause similar shifts, termed *wafer-induced shift* (WIS). Unlike TIS, however, WIS is rarely systematic enough to be simply subtracted out of the data. Additionally, nonideal targets often interact with the measurement tool, especially focus, to produce nonsystematic WIS errors.

Recently, a new type of target called an Advanced Imaging Metrology (AIM) target has shown superior measurement results (Figure 8.10). The AIM bars are less sensitive

to chemical mechanical polishing (CMP) errors such as dishing and other processing errors that can damage a box-in-box target, thus making the AIM target more process-robust. Also, the multiple bars essentially allow an averaging of multiple measurements at one time, significantly increasing the precision of the measurement. Special image processing software is used to relate the optical image of the target (determining the center of symmetry of each of the bars) to x - and y -overlay values.

Overlay measurement targets are placed at various spatial positions in the reticle field, minimally at the four corners but often in the streets between the die (and thus within the field) as well. The spatial variation of overlay errors provides extremely important information as to the behavior of the reticle, the wafer and the stepper, as we shall see in the next section. Overlay errors are characterized in terms of x -errors and y -errors separately, rather than being combined into an overlay error length. Since integrated circuit (IC) patterns are mostly laid out on a rectangular grid, the impact of an x -overlay error on device yield or performance is almost independent of any y -overlay errors that might also be present, or vice versa.

8.4.2 Analysis and Modeling of Overlay Data

The goal of overlay data analysis is twofold: assess the magnitude of overlay errors and determine, if possible, their root causes. The overall magnitude of overlay errors is commonly summarized by the mean plus 3σ of the overlay data in both x - and y -directions. Root cause analysis involves explaining the data with a model that assigns a cause to the observed effect.

To see how an overlay model might work, let's examine a simple example of a reticle that is slightly rotated. Figure 8.11a shows how a rectangular reticle field would print on a wafer if the reticle were rotated slightly with respect to the previous layer. If the ideal position of the upper-right corner of the field were (x,y) , rotation of the reticle about an angle ϕ would result in that corner printing at a location (x^*,y^*) given by

$$\begin{aligned}x^* &= x \cos \phi - y \sin \phi \\y^* &= y \cos \phi + x \sin \phi\end{aligned}\tag{8.18}$$

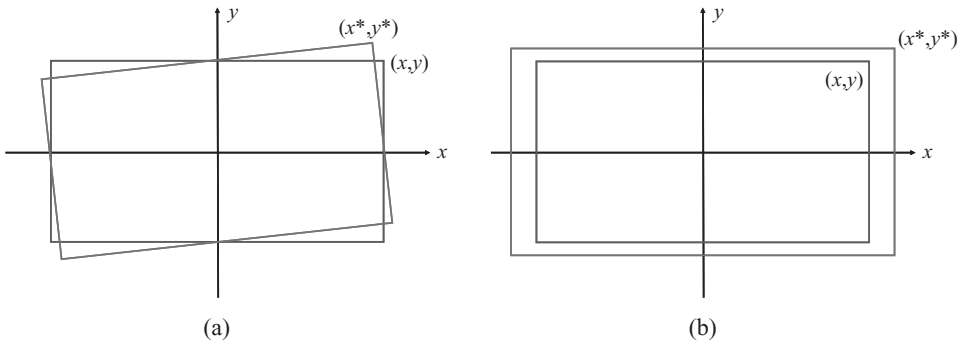


Figure 8.11 Examples of two simple overlay errors: (a) rotation and (b) magnification errors

where the origin was set to be in the middle of the field. In any reasonable lithography process, these rotational errors will be quite small (on the order of $10\ \mu\text{radians}$ or less) so that the trigonometric functions can accurately be replaced by their small angle approximations: $\cos\phi \approx 1$ and $\sin\phi \approx \phi$. The resulting overlay errors will then be

$$\begin{aligned} dx &\equiv x^* - x = -\phi y \\ dy &\equiv y^* - y = \phi x \end{aligned} \quad (8.19)$$

Another possible alignment error is translation (also called offset), where the entire reticle field is shifted in x and y by Δx and Δy . In this case, the resulting overlay errors are simply equal to this translation error:

$$\begin{aligned} dx &= \Delta x \\ dy &= \Delta y \end{aligned} \quad (8.20)$$

Another interesting error that can occur in optical lithography is a magnification error. While lithography tools are designed to reduce the reticle by a specific amount (most commonly $4\times$), in fact changes in the index of refraction of air and the exact position of the lens or lens components relative to the reticle and wafer can cause small errors in the magnification of the imaging tool. Sometimes these magnification errors can be different in the x - and y -directions (this is especially true for step-and-scan tools, where the ratio of reticle-to-wafer scan speeds controls the magnification in the scan direction). From Figure 8.11b, for x and y relative magnification errors of Δm_x and Δm_y , respectively,

$$\begin{aligned} x^* &= (1 + \Delta m_x) x \\ y^* &= (1 + \Delta m_y) y \\ dx &= \Delta m_x x \\ dy &= \Delta m_y y \end{aligned} \quad (8.21)$$

The uniform or overall magnification error is the average of Δm_x and Δm_y . The difference between Δm_x and Δm_y is sometimes called *anamorphism*.

Combining the three sources of overlay errors we've discussed so far, the resulting total error in overlay would be

$$\begin{aligned} dx &= \Delta x - \phi y + \Delta m_x x \\ dy &= \Delta y + \phi x + \Delta m_y y \end{aligned} \quad (8.22)$$

Like magnification, rotational errors are sometimes broken up into x and y rotations as well, with the uniform rotational error being the average of ϕ_x and ϕ_y and the difference between these angles called *skew* or *nonorthogonality*. If the overlay error (dx , dy) were measured at different points in the field [different (x,y) points], one could find the best fit of Equation (8.22) to the overlay data and extract the five unknown overlay error sources: rotation, x and y translation, and x and y magnification errors.

Is there a difference between a rotational error applied to the reticle and a rotational error applied to the wafer? Since the reticle field is repeated many times on one wafer, rotating the reticle is very different from rotating the wafer, as can be seen more clearly in Figure 8.12. (As an aside, the difference between wafer rotation in the x - and y -

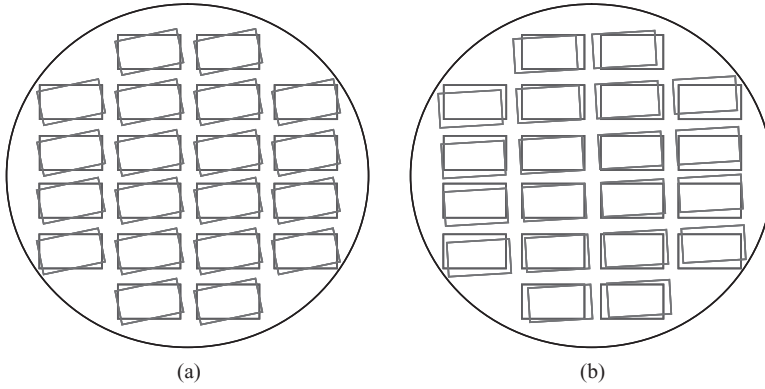


Figure 8.12 Different types of rotation errors as exhibited on the wafer: (a) reticle rotation and (b) wafer rotation

directions is called skew or orthogonality errors, and results from errors in the stepping motion of the wafer stage. That is why wafer errors are sometimes called stage errors.) Similarly, a reticle magnification error (caused by the imaging tool) would yield a different signature than a wafer magnification (also called scale) error (caused, for example, by a thermal expansion of the wafer). Translation errors, however, are exactly the same regardless of whether the reticle has been offset or the wafer has been offset.

Wafer rotation and magnification errors can be separated from reticle rotation and magnification errors by properly defining the coordinate axes used in Equation (8.22). If x and y represent the coordinates relative to the center of the field being measured, then the model fitting will yield reticle errors. If the coordinates are for the entire wafer (with an origin at the center of the wafer, for example), the resulting model fit will give wafer errors. By combining reticle and wafer terms in the full model (using both wafer and field coordinates in the full equation), both reticle and wafer error terms can be determined. Remember that translation errors cannot be separated into reticle and wafer, so only one set of translation terms should be used. The final model equations are

$$\begin{aligned} dx &= \Delta x - \theta_x y_{\text{wafer}} + \Delta M_x x_{\text{wafer}} - \phi_x y_{\text{field}} + \Delta m_x x_{\text{field}} \\ dy &= \Delta y + \theta_y x_{\text{wafer}} + \Delta M_y y_{\text{wafer}} + \phi_y x_{\text{field}} + \Delta m_y y_{\text{field}} \end{aligned} \quad (8.23)$$

where θ and ΔM represent the wafer (stage) rotation and magnification errors, and ϕ and Δm represent the reticle (field) rotation and magnification terms.

In the linear model developed here, there are a total of 10 model coefficients: two magnification and two rotation errors each for the reticle and the wafer, and two translation terms. A typical sample plan would measure overlay at the four corners of an exposure field for five different fields (in the middle and four ‘corners’ of the wafer), and for three wafers in a lot. This number of measurements is generally enough to get good modeling statistics, averaging random errors so that the model coefficients are reasonably precise. The residuals, the difference between the modeled overlay errors (the systematic errors) and the actual measured data, are a measure of the random component of the errors (Figure 8.13).

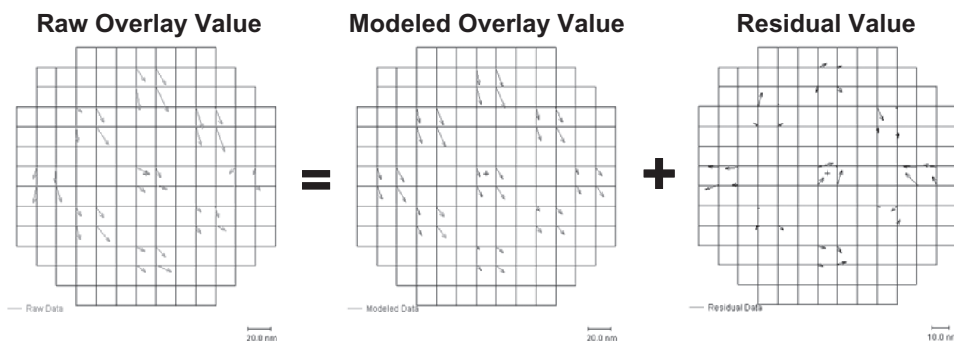


Figure 8.13 Separation of raw overlay data into modeled + residual values. The sampling shown here, four points per field and nine fields per wafer, is common for production monitoring

8.4.3 Improving Overlay Data Analysis

As discussed above, the primary overlay errors to consider are rotation of the wafer about an angle θ , rotation of the reticle about an angle ϕ , translation where the entire reticle field or wafer is shifted in x and y by Δx and Δy , relative wafer magnification errors of ΔM_x and ΔM_y in x and y , and relative field magnification errors of Δm_x and Δm_y in x and y , respectively. Combining these five sources of overlay errors gave the model in Equation (8.23). While this basic overlay model captures the majority of systematic errors typically found in stepper or step-and-scan imaging, additional model terms and other approaches can be used to improve the analysis. The goal is to properly account for all systematic overlay errors such that the model residuals, the difference between the best-fit model values and the actual data, are purely random. These additional terms are especially important when the first and second lithography levels are printed on different lithographic tools.

The linear terms in Equation (8.23) can be expanded to include higher-order terms. Second-order terms can model a trapezoidal error in the shape of the printed field, caused by a tilt in the reticle plane for nontelecentric imaging tools. Thus, these terms provide additional ‘correctables’ that can be used to adjust reticle tilt and reduce subsequent overlay errors. Third- and fifth-order terms are added to the field analysis to account for systematic lens distortion. While these terms are often not correctable, their inclusion in the model can improve the accuracy of the resulting lower-order model terms. A common higher-order model (including only the reticle field terms) is shown here:

$$\begin{aligned} dx &= -\phi_x y + \Delta m_x x + t_{1x} x^2 + t_{2x} xy + d_{3x} x(x^2 + y^2) + d_{5x} x(x^2 + y^2)^2 \\ dy &= \phi_y x + \Delta m_y y + t_{1y} y^2 + t_{2y} xy + d_{3y} y(x^2 + y^2) + d_{5y} y(x^2 + y^2)^2 \end{aligned} \quad (8.24)$$

where t_1 and t_2 are the trapezoidal error terms, d_3 are the third-order distortion terms, and d_5 represent fifth-order distortion. Figure 8.14 shows an overview of reticle model terms.

The purpose of using third- and fifth-order distortion terms in modeling overlay data is to account for the very real impact of systematic distortions caused by lens aberrations.

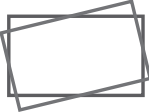
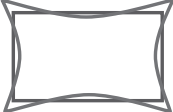
Error Term	Picture	Coefficients
Translation		$\Delta x, \Delta y$
Rotation		ϕ_x, ϕ_y
Magnification		$\Delta m_x, \Delta m_y$
Trapezoid (Keystone)		t_1, t_2
Lens Distortion		d_3, d_5

Figure 8.14 Illustration of field (reticle) model terms including higher-order trapezoid and distortion⁶

More correctly, such overlay errors are the result of *differences* in the distortion characteristics of the lenses used to print the first and second lithographic levels. If the same exposure tool was used for both levels, these high-order distortion terms should disappear (since lens distortion changes only slowly with time), regardless of the distortion of the lens relative to an absolute grid.

While modeling with high-order terms is a reasonable approach, it would be better to characterize the lens distortion more completely. Since distortion caused by lens aberrations varies slowly compared to the correctable error terms, it can be characterized infrequently with a thorough analysis. First, one exposure tool is picked as the reference or ‘golden’ tool. A ‘golden’ artifact wafer is printed on this tool and the resulting first-layer overlay marks are etched into the silicon. A test reticle is used that puts many, many overlay measurement targets throughout the exposure field (a 10×10 array, for example). Then, this artifact wafer is printed with the second layer on each exposure tool to be characterized. Measurement results are analyzed to subtract out the low-order correctables and multiple fields are measured to average out random errors. The remaining residuals form a *lens fingerprint* for each tool relative to the reference tool (Figure 8.15).

Once a database of lens fingerprints for each tool has been collected, these fingerprints can be used in different ways. In the simplest application, the lens fingerprint can be subtracted from the raw overlay data before modeling those data. Suppose stepper A was used as the reference and the current lot used stepper D for the first layer and stepper G for the second layer. The measured fingerprints in the database are D-A and G-A. The

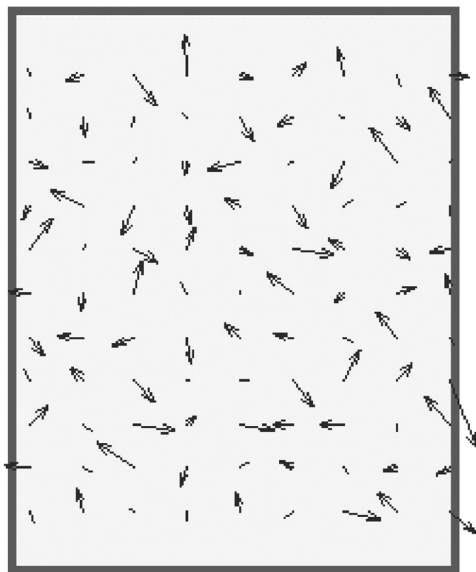


Figure 8.15 Example of a stepper lens fingerprint showing in this case nearly random distortion across the lens field

lens fingerprint for this lot would be G-A-D-A. In a second application of lens fingerprint data, every possible combination of steppers would be used to calculate the resulting fingerprints, which then are rank ordered by the maximum error in the fingerprint. This lens matching exercise results in a list of the best and worst combinations of steppers/scanners to use.

Like the imaging lens, every reticle contains fixed errors in registration (overlay relative to an absolute grid). Differences between the *reticle fingerprint* of the first and second lithography layer result in systematic overlay errors across the field that are independent of both lens distortion and stepper/scanner correctables. And just like the lens fingerprint, if this reticle fingerprint were to be measured, it could be used in the data analysis to improve the resulting modeling. Unfortunately, reticle fingerprints are much harder to measure and generally require special registration measurement equipment found only in the mask-making shop. Thus, measurement of a reticle fingerprint must generally be specified at the time of mask purchase.

Awareness of the reticle fingerprint makes measuring the lens fingerprint more difficult. If two reticles are used for the lens fingerprint measurement, the result will be a combined lens and reticle fingerprint measurement. Thus, typically one reticle is used for a lens fingerprint measurement, with both lithography pass patterns (inner and outer boxes, for example) located next to each other to minimize reticle effects.

The switch from pure scanning or pure stepping motion of the wafer stage to a combined step-and-scan stage motion adds several new possible overlay error terms. One such scanner term that affects the wafer overlay model is called *backlash*. With backlash, the x and y translation errors depend on the direction of the scan. Step-and-scan tools work

by scanning the wafer and reticle past the exposure slit until a whole field is exposed. Then the wafer is stepped to a new location and the scanning repeats in the opposite direction. Besides an overall translation error between reticle and wafer, backlash adds a translation error that switches sign depending on the direction of the scan. Backlash terms are often added to the standard overlay models [the 10-parameter model of Equation (8.23) becomes a 12-parameter model], and while most common for step-and-scan tools, it is sometimes used for steppers as well.

Scanning also adds the possibility of higher-order stage ‘wobble’. If a stage were scanning in the y -direction, for example, that stage could slowly wobble back and forth in the x -direction as it scans. Thus, a y -direction scan adds error terms to the x -overlay error dx that go as powers of the field coordinate y . A y^2 term is called *scan bow*. A y^3 term adds a snake-like signature to the field overlay. Note that a linear variation in y is already included as a reticle rotation.

Like the reticle and the lens, a specific scanner stage can have its own signature. However, unlike the reticle or the lens, the scanner stage signature varies more quickly over time than the other signatures. Thus, ideally each of these signatures should be measured independently. The lens fingerprint can be measured independent of the scan signature by using the lens fingerprint measurement method above for a static exposure (that is, with no scanning motion of stage and reticle). This lens signature within the slit can then be collapsed to a one-dimensional slit signature by averaging all the errors across the slit (that is, in the scan direction). Finally, a dynamic lens/scan signature can be measured by applying the lens fingerprint method for a full scanning exposure (see Figure 8.16). Subtracting out the slit signature leaves the scan signature remaining.

8.4.4 Using Overlay Data

Once overlay on a production lot of wafers has been measured and modeled, how is this data and analysis to be used? In general, production lot measurements of any kind can

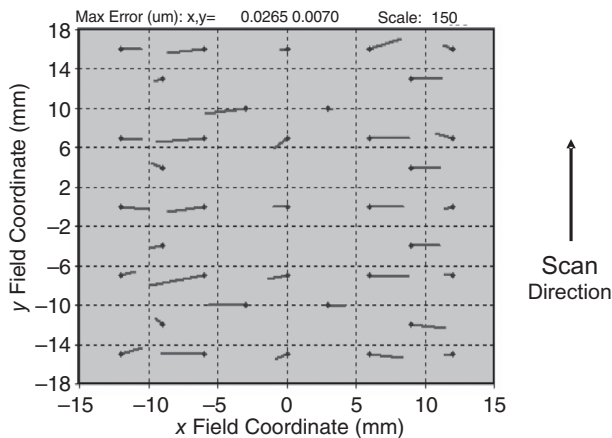


Figure 8.16 Example of a scanner lens/scan fingerprint⁷ (figure used with permission). Note that errors in the scan direction are mostly averaged out by the scan

be thought of as ‘health monitors’. The goal is to answer three questions: How healthy is this lot? How healthy are the process and process tools? How healthy is the metrology? For overlay, this translates into three basic uses of the metrology data: (1) lot dispositioning – should this lot be reworked or sent on to etch; (2) stepper correctables – have we detected errors in the stepper or scanner that can be adjusted for improved overlay; and (3) metrology diagnostics – do we have a problem with the targets, sample plan, or metrology tool that should be taken into account.

If overlay errors on a wafer are too large, final electric yield for that wafer can suffer. Since lithography offers the unique option to rework wafers that have not been properly processed, it is very cost-effective to try to catch yield-limiting overlay errors while the option for rework still exists. Since wafers are processed in batches called lots (typically made up of 25 wafers, though lot sizes can vary), and since each lot is processed in an essentially identical manner (common tool settings and materials, very small elapsed time between processing of wafers within a lot), these lots are almost always dispositioned together as a group.

After a lot has been processed through resist development, two or three wafers from that lot are randomly selected for overlay measurement. (For dual wafer stage scanners, care must be taken to sample both stages and analyze the stage/wafer errors separately.) Data analysis leads to a number of statistical and model-based metrics from those measurements. This collection of metrics must then be distilled into a binary go-no go decision: Do we predict that the cost of yield losses due to overlay errors, should the lot be passed, will exceed the cost of reworking the wafers? Some basic cost analysis can translate this question into a yield-loss threshold: if the predicted yield loss due to overlay errors exceeds a certain value, the lot should be reworked.

But standard overlay data analysis does not directly provide a prediction of overlay-limited yield. Instead, certain statistical or modeled proxies for yield are used. In the simplest approach, if the maximum measured error, the mean value of the error magnitude, or the standard deviation of the measured data distribution (in either x or y) exceed some predefined thresholds, the lot is to be reworked. These thresholds are generally determined by a combination of experience and simple rules of thumb that have been scaled with each technology node. Similarly, modeling results can also be used to assess the value of reworking the lot. If applying the modeled correctables would reduce overlay errors for this lot by some predefined amount, it becomes worthwhile to rework.

The above approaches for assessing yield have a serious flaw – they only predict the impact on yield of the errors that occur at the measurement points. But measurement sampling on production lots is generally sparse to keep measurement costs low. Thus, it is most likely that errors at nonsampled points will be the yield limiters. Modeling can be used to overcome this limitation by predicting the maximum error expected on the wafer. The best-fit model can be used to predict the systematic variation of overlay at any point on the wafer. Using the residuals of the model fit as an estimate of the random error contribution, a maximum predicted error can be calculated for an average field and for the wafers. If this maximum predicted error is above some threshold, the lot is reworked.

Even the more rigorous approach of calculating the predicted maximum error on the wafer has a flaw: wafers don’t fail due to overlay errors, die fail. In general, each exposure field contains several die (2–4 microprocessors or 8–16 memory chips in a field are

common). Knowing the number of die per field and the pattern of fields printed on the wafer, the max predicted error approach can be used to find the maximum error that occurs in each die on the wafer. By predicting that a catastrophic yield-loss error will occur whenever the overlay exceeds some threshold at any point in the die, the max predicted error calculation can be used to predict the overlay-limited yield – the number of die that are expected to fail due to overlay errors from the current lithography step. This overlay-limited yield metric translates measured overlay data into a lot pass/fail decision that is directly related to final die yield.

As discussed before, one important use of overlay data modeling is to calculate stepper correctables. Correctables are defined as corrections to the exposure tool settings (affecting reticle and wafer position and motion) that, if they had been made before the measured lot was processed, would have resulted in improved overlay for that lot. Thus, technically these correctables are backward-looking. But their true value comes from applying them in a forward-looking manner. There are several ways in which overlay correctables are used. The most direct and obvious is during lot rework: if the measurements indicate that the lot should be reworked, what stepper/scanner settings should be changed to best ensure reworking will result in significantly improved overlay? Since reworking is typically done immediately after that lot was first processed, without unloading the reticle or changing the basic settings of the tool, the correctables tend to work extremely well at improving the reworked overlay.

Correctables can also be applied even if the current lot is passed. Often an exposure tool is used to process numerous identical lots in a row. Like the rework case, the reticle is not being reloaded and the basic stepper settings are not being changed between lots. Thus, correctables can be used to keep the overlay errors of each subsequent lot at their minimum. A system of feeding back correctables based on current measurements to keep the next batch of wafers in control is termed APC, *advanced process control*. Finally, correctables can be tracked over time for a given exposure tool to look for trends. SPC (statistical process control) techniques can determine if the variation in correctables from lot to lot is statistically unexpected and thus cause for alarm.

The third ‘health monitor’ use of production lot overlay measurements is to diagnose the health of the metrology itself. Overlay measurement begins with a signal acquisition, which is then analyzed to produce the raw overlay data. The signal can also be analyzed for noise and nonideal characteristics (such as asymmetries) in what is called *target diagnostics*. Wafer processing, film deposition and chemical mechanical polishing (CMP) in particular, can reduce the visibility of target to an optical measurement tool both decreasing the signal and increasing the noise. CMP and its variation across the wafer can also cause an asymmetric ‘smearing’ of the topography that makes up the overlay measurement target.

Target diagnostics can be used to assess the uncertainty of a given measurement. This uncertainty can be used to weight the data for subsequent modeling, or as a data removal mechanism (if the target is too bad, measurement results from that target can be ignored). But the modeling itself offers another diagnostic tool for the metrology. Fitting the data to a model results in model coefficients plus residuals. If the residuals are too high, this could be an indication that something has gone wrong with the measurement. Additionally, the data fitting process results not only in model coefficients but in estimates of the uncertainties of those coefficients (technically termed the covariance matrix of the fit). If

the uncertainty of a correctable is consistently too high, there is a good chance that the sampling plan needs to be improved.

8.4.5 Overlay Versus Pattern Placement Error

Overlay is measured using special measurement targets which, in general, are large compared to the resolution limits of the lithographic imaging tool. As was discussed in Chapter 3, however, aberrations can cause errors in the positions of printed features that are a function of the feature size and pitch of the patterns. The positional errors specific to the feature being printed are called *pattern placement errors* (PPE) and are above and beyond (and thus can be added directly to) the overlay errors that are the same for any pattern. Using the Zernike polynomial terminology, the variation of x - and y -tilt with field position, called distortion, is an overlay error since the same error is experienced by every feature. However, the higher-order odd aberrations (such as coma) cause additional positional errors that depend on the feature. These additional errors are called pattern placement errors.

Pattern placement errors can be measured by using special overlay targets that have the same (or nearly the same) structure as the device patterns of interest. Since practically this can be very difficult to accomplish, pattern placement errors are more commonly predicted by calculating the various images of interesting patterns given measured lens aberrations as a function of field position (see Chapter 3).

8.5 The Process Window

As discussed in section 8.2.2, there are two ways to improve CD control: reduce the variation in CD with a given process error (that is, increase process latitude); and reduce the magnitude of process errors. In order to increase process latitude, one must first measure and characterize that process latitude. Unfortunately, there are a very large number of potential process errors in the fab, from variations in the wafer filmstack to batch-to-batch variations in resist properties, from scanner stage vibrations to PEB hot plate temperature nonuniformities. To fully characterize the response of CD to each individual process error seen in manufacturing would be a daunting task, let alone to consider the interactions of these errors.

Fortunately, the task is simplified by recognizing that essentially all errors in the fab fall into two basic categories: errors that behave like dose errors, and errors that behave like focus errors. Once this bifurcation of error sources is understood, the job of measuring process latitudes is simplified to measuring the focus and dose responses of the process. The sections below will address the importance of dose and focus by explaining the *focus-exposure matrix*, and using it to provide definitions of the *process window* and *depth of focus* (DOF).

8.5.1 The Focus-Exposure Matrix

In general, depth of focus can be thought of as the range of focus errors that a process can tolerate and still give acceptable lithographic results. Of course, the key to a good definition of DOF is in defining what is meant by tolerable. A change in focus results in

two major changes to the final lithographic result: the photoresist profile changes, and the sensitivity of the process to other processing errors is increased. Typically, photoresist profiles are described using three parameters: the linewidth (or critical dimension, CD), the sidewall angle and the final resist thickness (that is, the profile is modeled as a trapezoid, see section 8.2.4). The variation of these parameters with focus can be readily determined for any given set of conditions (although CD is the easiest of the three to measure). The second effect of defocus is significantly harder to quantify: as an image goes out of focus, the process becomes more sensitive to other processing errors such as exposure dose, PEB temperature and develop time. Exposure dose is commonly chosen to represent these other process responses.

The effect of focus on a resist feature is dependent on exposure. Thus, the only way to judge the response of the process to focus is to simultaneously vary both focus and exposure in what is known as a *focus–exposure matrix*. Figure 8.17 shows a typical example of the output of a focus–exposure matrix using linewidth as the response (sidewall angle and resist loss can also be plotted in the same way) in what is called a Bossung plot.⁸

Unfortunately, experimental data collected from a focus–exposure matrix will rarely be as smooth and clean as that shown in Figure 8.17. Experimental error is typically several percent for CDs at best focus and exposure, but the error grows appreciably when out of focus. To better analyze focus–exposure CD data, it is common to fit the data to a reasonable empirical equation in order to reduce data noise and eliminate flyers. A reasonable expression can be derived from a basic understanding of how an image is printed in photoresist.

Consider an ideal, infinite contrast positive resist and simple 1D patterns. For such a resist, any exposure above some threshold dose E_{th} will cause all of the resist to be removed (see Chapter 3). For an aerial image $I(x)$ and an exposure dose E , the CD will equal $2x$ when

$$E_{th} = EI(x) \quad (8.25)$$

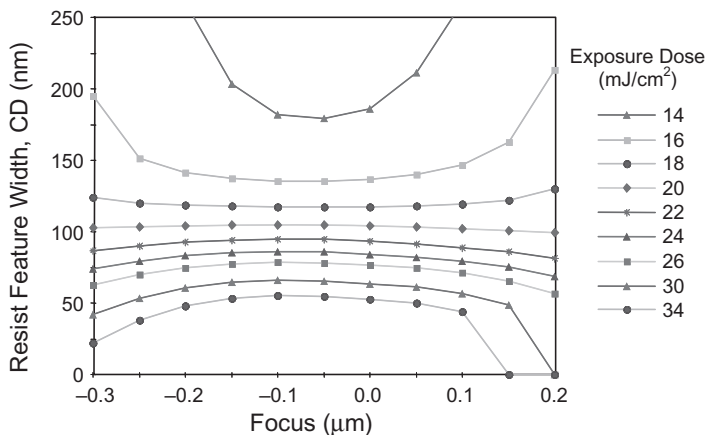


Figure 8.17 Example of the effect of focus and exposure on the resulting resist linewidth. Here, focal position is defined as zero at the top of the resist with a negative focal position indicating that the plane of focus is inside the resist

Consider some reference dose E_1 producing $x_1 = CD_1/2$. For some small difference from this x value, let us assume the aerial image is linear (that is, let us use only the first two terms in a Taylor series expansion of I about x_1).

$$I(x) = I(x_1) + \frac{dI}{dx}(x - x_1) \quad (8.26)$$

Combining Equations (8.25) and (8.26),

$$E_{th} = E_1 I(x_1) = EI(x) = E \left[I(x_1) + \frac{dI}{dx}(x - x_1) \right] \quad (8.27)$$

Dividing both sides of the equation by $EI(x_1)$,

$$\frac{E_1}{E} = \frac{I(x)}{I(x_1)} = 1 + \frac{d \ln I}{dx}(x - x_1) \quad (8.28)$$

where the log-slope of the aerial image is evaluated at x_1 . Since $x = CD/2$, this equation can be rearranged to give

$$\frac{E_1}{E} - 1 = \frac{d \ln I}{d \ln CD} \left(\frac{CD - CD_1}{CD_1} \right) \quad (8.29)$$

Since we have assumed an infinite contrast resist, it is easy to show from Equation (8.25) that

$$\frac{d \ln I}{d \ln CD} = - \frac{d \ln E}{d \ln CD} \quad (8.30)$$

Combining Equations (8.29) and (8.30),

$$\frac{CD - CD_1}{CD_1} = \frac{d \ln CD}{d \ln E} \bigg|_{E=E_1} \left(1 - \frac{E_1}{E} \right) \quad (8.31)$$

The term $d \ln CD / d \ln E$ is the slope of the CD-versus-dose curve when plotted on a log-log scale, evaluated at the reference dose E_1 . Thus, from this simple model we have a prediction of how CD varies with dose.

An alternate exposure dependence can be derived by assuming the log-slope of the aerial image is constant around x_1 rather than the slope of the image. In this case, Equation (8.26) is replaced with

$$I(x) = I(x_1) e^{s(x-x_1)} \quad \text{where} \quad s = \frac{d \ln I}{dx} \bigg|_{x=x_1} \quad (8.32)$$

Following the same derivation path as before,

$$\frac{CD - CD_1}{CD_1} = \frac{d \ln CD}{d \ln E} \bigg|_{E=E_1} \ln \left(\frac{E}{E_1} \right) \quad (8.33)$$

Note that Equation (8.33) reverts to Equation (8.31) with a Taylor expansion of the natural logarithm when E is near E_1 .

Finally, the most accurate aerial image dose variation model assumes the log-slope of the image varies linearly with x . As was discussed in Chapter 7, this leads to an aerial image of the space with a Gaussian shape:

$$I(x) = I_0 e^{-NLS_1(x/CD_1)^2} \quad \text{where} \quad NLS_1 = CD_1 \left. \frac{d \ln I}{dx} \right|_{x=x_1} \quad (8.34)$$

Proceeding as before,

$$\frac{CD}{CD_1} = \sqrt{1 + 2 \left. \frac{d \ln CD}{d \ln E} \right|_{E=E_1} \ln \left(\frac{E}{E_1} \right)} \quad (8.35)$$

where in this case the CD is for the space. For a line, the equivalent expression for CD would be

$$\frac{p - CD}{p - CD_1} = \sqrt{1 - 2 \left(\frac{CD_1}{p - CD_1} \right) \left. \frac{d \ln CD}{d \ln E} \right|_{E=E_1} \ln \left(\frac{E}{E_1} \right)} \quad (8.36)$$

where p is the pitch. Note that a Taylor series expansion of the square root when E is near E_1 leads to Equation (8.33). Equation (8.34) for a Gaussian-shaped image has its center at $x = 0$, the center of the space. This describes typical aerial images very well for small spaces, but for larger spaces there will be an offset between the peak of the Gaussian and the center of the space. In this case, Equation (8.36) can still be used if the pitch is replaced by an empirically determined effective pitch.

Equation (8.31), (8.33) or (8.35) describes how CD varies with dose for any given aerial image [though in practice Equation (8.33) doesn't work nearly as well as Equation (8.31) or (8.35), and Equation (8.35) matches data best of all]. Thus, these expressions apply to an out-of-focus image just as well as to an in-focus image. The only thing that changes through focus is the CD -versus-dose slope. To first order, this slope varies quadratically with focus.

$$\left. \frac{d \ln CD}{d \ln E} \right|_{F=F_0} = \left. \frac{d \ln CD}{d \ln E} \right|_{F=F_0} \left[1 + \left(\frac{F - F_0}{\Delta} \right)^2 \right] \quad (8.37)$$

where F is the focal position, F_0 is best focus, and Δ is the focus error that causes the log- CD -versus-log-dose slope to double.

As an example, consider the case of pure three-beam coherent imaging of equal lines and spaces through focus as discussed in Chapter 3. The aerial image was shown to be

$$I(x) = \frac{1}{4} + \frac{2}{\pi^2} + \frac{2}{\pi} \cos(\Delta\Phi) \cos(2\pi x/p) + \frac{2}{\pi^2} \cos(4\pi x/p) \quad (8.38)$$

where $\Delta\Phi = 2\pi n(F - F_0) \left(1 - \sqrt{1 - (\lambda/p)^2} \right) / \lambda$.

The normalized log-slope of the image evaluated at the nominal line edge is

$$NLS = 8 \cos(\Delta\Phi) \quad (8.39)$$

Recalling Equation (8.30) and assuming that the defocus amount is small enough that the cosine can be replaced by the first two terms of its Taylor series expansion,

$$\frac{d \ln CD}{d \ln E} = \frac{d \ln CD}{d \ln E} \bigg|_{F=F_0} \left[1 + \frac{\Delta \Phi^2}{2} \right] \quad (8.40)$$

This gives the same result as Equation (8.37) with

$$\Delta = \frac{\lambda}{\sqrt{2}\pi n} \left(1 - \sqrt{1 - (\lambda/p)^2} \right)^{-1} \approx \frac{\sqrt{2}p^2}{\pi n \lambda} \quad (8.41)$$

This method of adding the focus dependence to the dose dependence gives extra significance to the dose E_1 . When $E = E_1$, Equations (8.31), (8.33) and (8.35) require that $CD = CD_1$ regardless of the value of focus. As will be discussed in section 8.5.3, this is the definition of the *isofocal dose* and *isofocal CD*. Thus, the exposure latitude term $d \ln CD / d \ln E$ used in the above expressions will be the value of this slope at the isofocal dose (see Figure 8.18). At other exposure energies, this log-CD-versus-log-dose slope will of course be different. For the simplest model, Equation (8.31), the result is

$$\frac{d \ln CD}{d \ln E} = \frac{d \ln CD}{d \ln E} \bigg|_{E=E_1} \left(\frac{E_1}{E} \right) \left(\frac{CD_1}{CD} \right) \quad (8.42)$$

For the more exact model of Equations (8.35) or (8.36),

$$\begin{aligned} \text{Space: } \frac{d \ln CD}{d \ln E} &= \frac{d \ln CD}{d \ln E} \bigg|_{E=E_1} \left(\frac{CD_1}{CD} \right)^2 \\ \text{Line: } \frac{d \ln CD}{d \ln E} &= \frac{d \ln CD}{d \ln E} \bigg|_{E=E_1} \left(\frac{CD_1}{CD} \right) \left(\frac{p - CD_1}{p - CD} \right) \end{aligned} \quad (8.43)$$

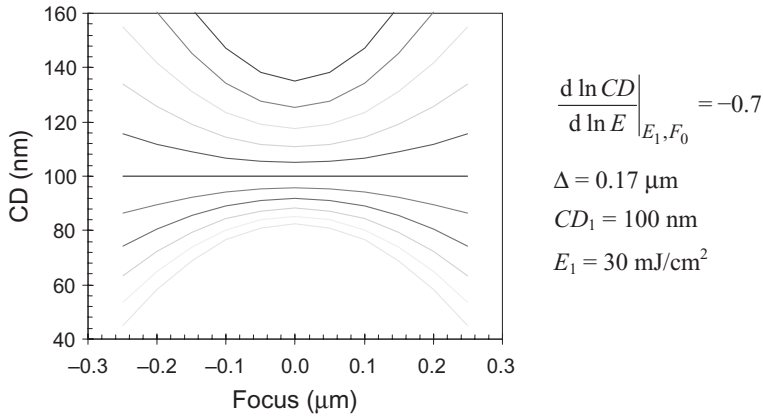


Figure 8.18 Plot of the simple Bossung model of Equations (8.31) and (8.37) shows that it describes well the basic behavior observed experimentally

Over reasonable ranges of focus (the magnitudes of focus errors normally encountered in semiconductor lithography), the impact of focus on the blurring of an aerial image is symmetric. That is, going out of focus in each direction (the wafer too close or too far from the projection lens) produces the same result on the aerial image for the same magnitude of focus error. And yet, when we measure the lithographic response to focus, in the form of a focus–exposure matrix or Bossung plot, we can observe an obvious asymmetry in the response of critical dimension (CD) to focus. Figure 8.17 shows a small but noticeable tilt in the shape of the Bossung curves and a decided asymmetry between plus and minus focus.

Since the aerial image responds to focus symmetrically, the cause of the observed asymmetry must involve the way that the aerial image interacts with the photoresist to produce the final CD. In fact, it is the thickness of the photoresist that gives rise to this asymmetry. As the aerial image travels through space, it becomes more ‘in focus’ as it approaches the focal plane, then defocuses as it travels away from this plane of best focus. Because the photoresist has a nonzero thickness, different parts of the resist will experience different amounts of defocus. Suppose the focal plane were placed just above the top of the resist (Figure 8.19a). At the top, the resist sees an aerial image that is near best focus and thus very sharp. As this image propagates toward the bottom of the resist, however, it goes further out of focus. As a consequence, the top of the resist profile looks sharp but the bottom is curved or ‘blurred’ by the defocused image. If, on the other hand, the plane of best focus were placed below the bottom of the resist (Figure 8.19b), a very different resist shape will result. The bottom of the resist will see a sharp image, while the top of the resist will experience a more defocused image. The result will be a resist profile with straight sidewalls at the bottom, but a very rounded top.

As Figure 8.19 shows, the shape of the resist profile will be very different depending on the direction of the defocus (note that ‘best’ focus will typically place the focal plane somewhere near the middle of the resist thickness). As one might expect, the critical dimensions of the two resist profiles shown in Figure 8.19 will also be different. The result is an asymmetric response to focus as shown in the Bossung curves. The asymmetry becomes apparent whenever the thickness of the photoresist becomes an appreciable fraction of the depth of focus of that feature.

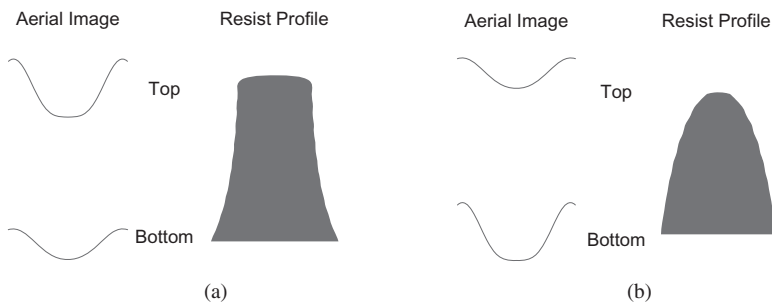


Figure 8.19 Positioning the focal plane (a) above the top of the resist, or (b) below the bottom of the resist results in very different shapes for the final resist profile

Asymmetric response to focus can be added to the semiempirical Bossung model by allowing other powers of dose and focus (in particular, allowing odd powers of focus). Thus, a generalized version of the Bossung model becomes

$$\frac{CD - CD_1}{CD_1} = \sum_{n=1}^N \sum_{m=0}^M a_{nm} \left(1 - \frac{E_1}{E}\right)^n (F - F_0)^m \quad (8.44)$$

In general, very good fits to experimental data are obtained with very few terms of the summation.⁹

8.5.2 Defining the Process Window and DOF

Of course, one output as a function of two inputs can be plotted in several different ways. For example, the Bossung curves could also be plotted as exposure latitude curves (linewidth versus exposure) for different focus settings. Another very useful way to plot this two-dimensional data set is a contour plot – contours of constant linewidth versus focus and exposure (Figure 8.20a).

The contour plot form of data visualization is especially useful for establishing the limits of exposure and focus that allow the final image to meet certain specifications. Rather than plotting all of the contours of constant CD, one could plot only the two CDs corresponding to the outer limits of acceptability – the CD specifications. Because of the nature of a contour plot, other variables can also be plotted on the same graph. Figure 8.20b shows an example of plotting contours of CD (nominal $\pm 10\%$), 80° sidewall angle and 10% resist loss all on the same graph. The result is a *process window* – the region of focus and exposure that keeps the final resist profile within all three specifications.

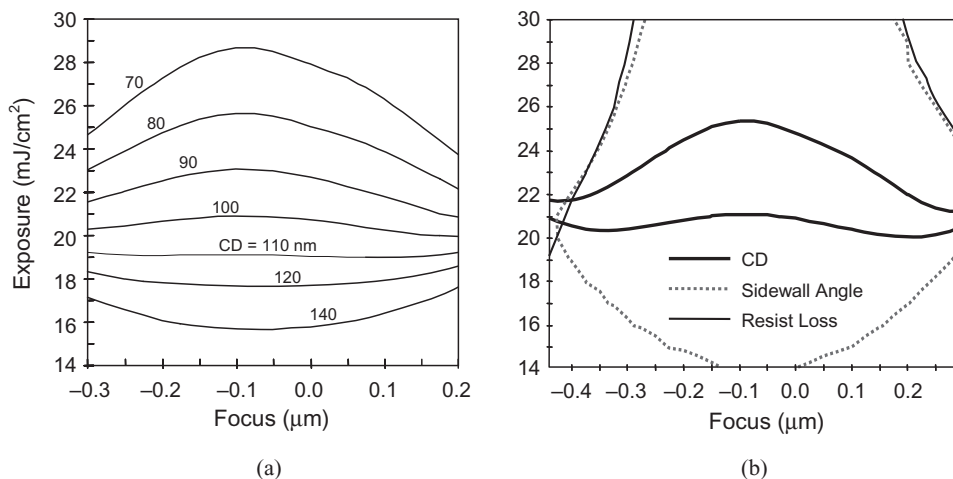


Figure 8.20 Displaying the data from a focus–exposure matrix in an alternate form: (a) contours of constant CD versus focus and exposure, and (b) as a focus–exposure process window constructed from contours of the specifications for linewidth, sidewall angle and resist loss

The focus–exposure process window is one of the most important plots in lithography since it shows how exposure and focus work together to affect linewidth (and possibly sidewall angle and resist loss as well). The process window can be thought of as a *process capability* – how the process responds to changes in focus and exposure. An analysis of the error sources for focus and exposure in a given process will give a *process requirement* (see Table 8.2). If the process capability exceeds the process requirements, yield will be high. If, however, the process requirement is too large to fit inside the process window, yield will suffer.

What is the maximum range of focus and exposure (that is, the maximum process requirement) that can fit inside the process window? A simple way to investigate this question is to graphically represent errors in focus and exposure as a rectangle on the same plot as the process window. The width of the rectangle represents the built-in focus errors of the processes, and the height represents the built-in dose errors. The problem then becomes one of finding the maximum rectangle that fits inside the process window. However, there is no one answer to this question. There are many possible rectangles of different widths and heights that are ‘maximal’, i.e. they cannot be made larger in either direction without extending beyond the process window (Figure 8.21a). (Note that the concept of a ‘maximum area’ is meaningless here.) Each maximum rectangle represents one possible trade-off between tolerance to focus errors and tolerance to exposure errors. Larger DOF can be obtained if exposure errors are minimized. Likewise, exposure latitude can be improved if focus errors are small. The result is a very important trade-off between exposure latitude and DOF.

If all focus and exposure errors were systematic, then the proper graphical representation of those errors would be a rectangle. The width and height would represent the total ranges of the respective errors. If, however, the errors were randomly distributed, then a probability distribution function would be needed to describe them. It is common to assume that random errors in exposure and focus are caused by the summation of many small sources of error, so that by the central limit theorem the overall probability

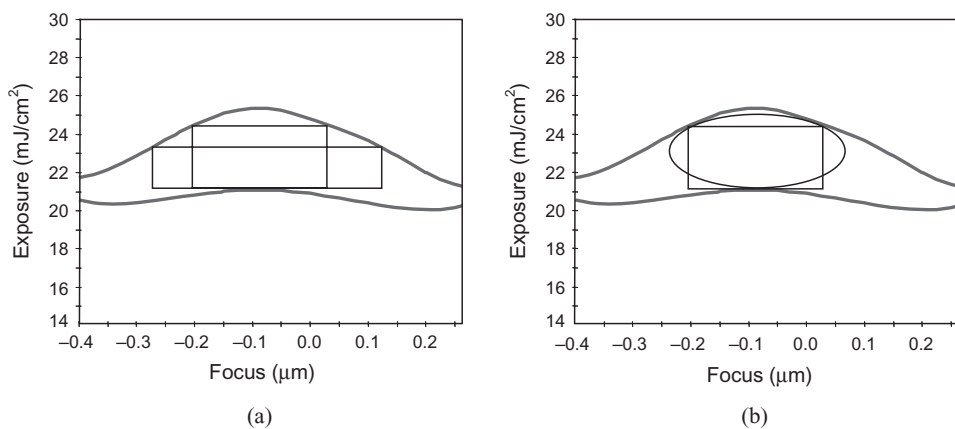


Figure 8.21 Measuring the size of the process window: (a) finding maximum rectangles; and (b) comparing a rectangle to an ellipse

distributions for focus and dose will be Gaussian (a normal distribution). Given standard deviations in exposure and focus, σ_E and σ_F respectively, the probability density function will be

$$p(\Delta E, \Delta F) = \frac{1}{2\pi\sigma_E\sigma_F} \exp(-\Delta E^2/2\sigma_E^2) \exp(-\Delta F^2/2\sigma_F^2) \quad (8.45)$$

where focus errors and exposure errors are assumed to be independent. In order to graphically represent the errors of focus and exposure, one should describe a surface of constant probability of occurrence. All errors in focus and exposure inside the surface would have a probability of occurring which is greater than the established cutoff. What is the shape of such a surface? For fixed systematic errors, the shape is a rectangle. For a Gaussian distribution, the surface can be derived by setting the probability of Equation (8.45) to a constant, p^* .

$$p^* = \frac{1}{2\pi\sigma_E\sigma_F} \exp(-\Delta E^2/2\sigma_E^2) \exp(-\Delta F^2/2\sigma_F^2) \quad (8.46)$$

$$-\ln(2\pi\sigma_E\sigma_F p^*) = \frac{\Delta E^2}{2\sigma_E^2} + \frac{\Delta F^2}{2\sigma_F^2}$$

Equation (8.46) is that of an ellipse. Suppose, for example, that one wishes to describe a 'three-sigma' surface, where p^* corresponds to the probability of having an error equal to 3σ in one variable. The resulting surface would be an ellipse with major and minor axes equal to $3\sigma_E$ and $3\sigma_F$ (Figure 8.21b).

$$1 = \frac{\Delta E^2}{(3\sigma_E)^2} + \frac{\Delta F^2}{(3\sigma_F)^2} \quad (8.47)$$

As seen in Table 8.2 for focus errors, the errors that actually occur in manufacturing are a combination of random and systematic errors. In fact, it is a reasonable rule of thumb to say that the full-range systematic errors are about equal in magnitude to (on the same order as) 6 times the standard deviation of the random errors. As a result, the total error probability distribution will be a complimentary error function (the random error Gaussian convolved with the systematic error step function). Since the tail of an error function has a shape very much like a Gaussian, the resulting surface of constant probability for small probabilities (like a 2σ or 3σ probability) will be very similar to an ellipse.

Using either a rectangle for systematic errors or an ellipse for random errors, the size of the errors that can be tolerated for a given process window can be determined. Taking the rectangle as an example, one can find the maximum rectangle that will fit inside the process window. Figure 8.22 shows an analysis of the process window where every maximum rectangle is determined and its height (the exposure latitude) plotted versus its width (depth of focus). Likewise, assuming Gaussian errors in focus and exposure, every maximum ellipse that fits inside the process window can be determined. The horizontal width of the ellipse would represent a three-sigma error in focus, while the vertical height of the ellipse would give a three-sigma error in exposure. Plotting the height versus the width of all the maximum ellipses gives the second curve of exposure latitude versus DOF in Figure 8.22.

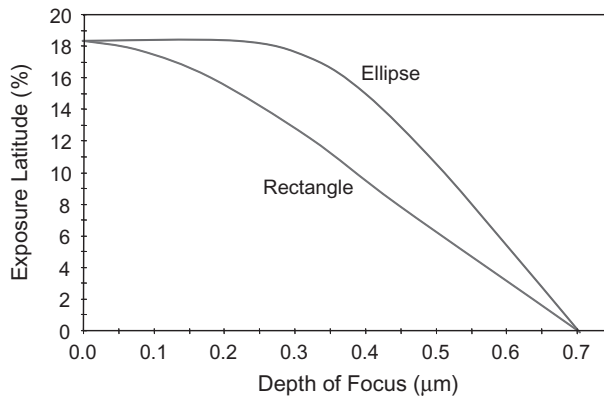


Figure 8.22 The process window of Figure 8.20b is analyzed by fitting all the maximum rectangles and all the maximum ellipses, then plotting their height (exposure latitude) versus their width (depth of focus)

The exposure latitude-versus-DOF curves of Figure 8.22 provide the most concise representation of the coupled effects of focus and exposure on the lithography process. Each point on the exposure latitude-DOF curve is one possible operating point for the process. The user must decide how to balance the trade-off between DOF and exposure latitude. One approach is to define a minimum acceptable exposure latitude, and then operate at this point; this has the effect of maximizing the DOF of the process. In fact, this approach allows for the definition of a single value for the DOF of a given feature for a given process. The depth of focus of a feature can be defined as *the range of focus that keeps the resist profile of a given feature within all applicable profile specifications (linewidth, sidewall angle and resist loss, for example) over a specified exposure range*. For the example given in Figure 8.22, a minimum acceptable exposure latitude of 10%, in addition to the other profile specifications, would lead to the following depth of focus results:

- DOF (rectangle) = $0.40\ \mu\text{m}$
- DOF (ellipse) = $0.52\ \mu\text{m}$

As one might expect, systematic errors in focus and exposure are more problematic than random errors, leading to a smaller DOF. Most actual processes would have a combination of systematic and random errors. Thus, one might expect the rectangle analysis to give a pessimistic value for the DOF, and the ellipse method to give an optimistic view of DOF. As mentioned above, however, the ellipse method gives a DOF very close to the true value for a common mix of systematic and random errors.

The definition of depth of focus also leads naturally to the determination of best focus and best exposure. The DOF value read off from the exposure latitude-versus-DOF curve corresponds to one maximum rectangle or ellipse that fits inside the process window. The center of this rectangle or ellipse would then correspond to best focus and exposure for this desired operating point. Knowing the optimum focus and dose values is essential to being able to use the full process window. If the process focus and dose settings deviate

from this optimum, the range of focus and dose errors that can be tolerated will be reduced accordingly. For example, if focus is set in error from best focus by an amount Δf , the resulting DOF will generally be reduced by $2\Delta f$.

8.5.3 The Isofocal Point

An interesting effect is observed in many focus–exposure data sets. There is often one exposure dose that produces a CD that has almost no sensitivity to focus. In the Bossung plot, this corresponds to the curve with the flattest response to focus over the middle of the focus range (Figure 8.23). By plotting CD through dose for different focuses, this isofocal point becomes even clearer as the point where the various curves cross. The dose that produces this isofocal behavior is called the *isofocal dose*, and the resulting CD is called the *isofocal CD*.

Ideally, the isofocal CD would equal the target CD of the process. Unfortunately, this is rarely the case without specifically designing the process for that result. The difference between the isofocal CD and the target CD is called the *isofocal bias*. A process with a large isofocal bias will produce smaller usable process windows because the right and left sides of the process window will bend either up or down, making the maximum possible ellipse or rectangle much smaller. In general, more isolated features exhibit greater isofocal bias than dense features, as seen in Figure 8.24.

As was mentioned in section 8.5.1, the isofocal CD and the isofocal dose play an important role in the semiempirical Bossung model of Equation (8.44). The terms E_1 and CD_1 are in fact identical to the isofocal dose and CD. For the simplest version of this model, Equations (8.31) and (8.37), the role of the isofocal point in influencing the shape of the process window can be made explicit. Since the model takes the form

$$\frac{CD - CD_1}{CD_1} = s \left(1 - \frac{E_1}{E} \right) \left[1 + \left(\frac{F - F_0}{\Delta} \right)^2 \right], \quad \text{where } s = \frac{d \ln CD}{d \ln E} \Big|_{E_1, F_0} \quad (8.48)$$

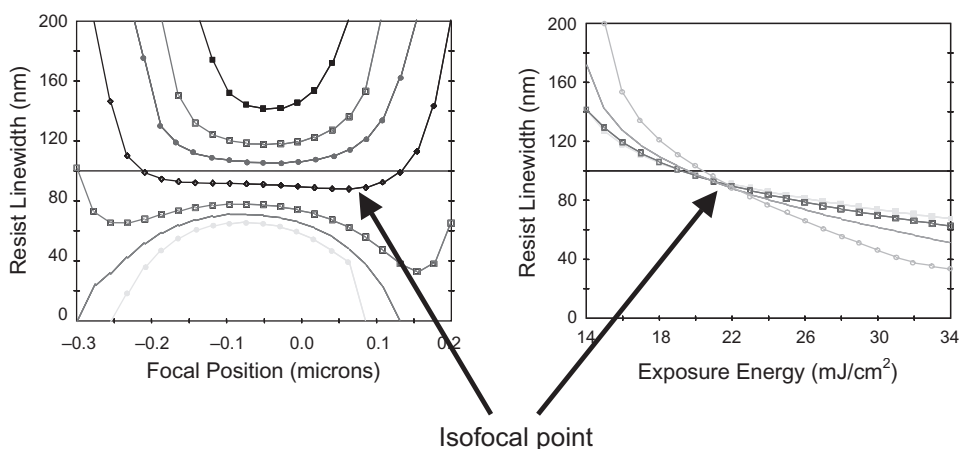


Figure 8.23 Two ways of plotting the focus–exposure data set showing the isofocal point – the dose and CD that have minimum sensitivity to focus changes (for the left graph, each curve represents a different exposure dose; for the right graph each curve is for a different focus)

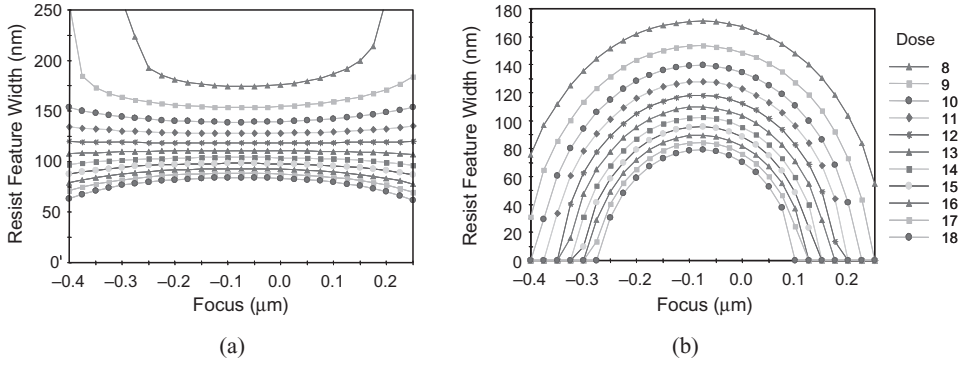


Figure 8.24 Bossung plots for (a) dense and (b) isolated 130-nm lines showing the difference in isofocal bias

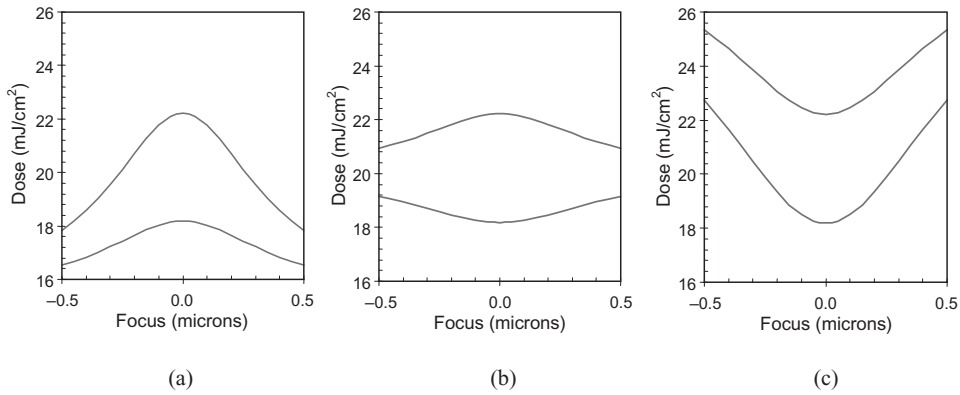


Figure 8.25 Process windows calculated using Equation (8.49) for a nominal line CD of 100 ± 10 nm, nominal dose of 20 mJ/cm^2 , $s = -1$, and $\Delta = 0.45$ microns: (a) isofocal $CD_1 = 130$ nm, (b) isofocal $CD_1 = 100$ nm and (c) isofocal $CD_1 = 70$ nm

contours of constant CD can be found by solving for $E(F, CD)$:

$$E(F, CD) = E_1 \frac{1 + \left(\frac{F - F_0}{\Delta} \right)^2}{1 + \left(\frac{F - F_0}{\Delta} \right)^2 - \left(\frac{CD - CD_1}{sCD_1} \right)^2} \quad (8.49)$$

(It is useful to remember that s will be negative for a line pattern, and positive for a space.) Figure 8.25 shows three example process windows, showing how the shape of the process window changes depending on whether the nominal CD is smaller, the same, or larger than the isofocal CD.

The process window defined by two contours as given by Equation (8.49) can be analytically measured to give the depth of focus. Using the rectangle approach to measuring

the size of the process window, consider first the case where the isofocal CD is within the process window (such as Figure 8.25b). The rectangle will be limited by its four corners, and the resulting DOF (for either a line or a space) will be approximately

$$DOF = 2\Delta \sqrt{\frac{CD_{\text{spec}}}{|s_{\text{nom}}| EL_{\text{spec}}} - 1} \quad (8.50)$$

where CD_{spec} is the range of acceptable CDs divided by the nominal CD (equal to 0.2 for the typical $\pm 10\%$ spec), EL_{spec} is the exposure latitude spec (for example, 0.1 for a 10% exposure latitude specification), and s_{nom} is the slope of the log-CD-versus-log-dose curve evaluated at the nominal dose [which can be related to s by Equation (8.42)]. As expected, the DOF can be improved with a larger Δ , a smaller s_{nom} , and relaxed CD and exposure latitude specifications.

When the isofocal CD is not within the process window (Figure 8.25a and c), the bending of the process window further limits the maximum rectangle that can fit inside the window and thus the DOF. For these cases, the DOF becomes

$$DOF \approx \frac{1}{K} 2\Delta \sqrt{\frac{CD_{\text{spec}}}{|s_{\text{nom}}| EL_{\text{spec}}} - 1} \quad (8.51)$$

where

$$K = \sqrt{\frac{CD_{\text{rel-iso-limit}}}{|s_{\text{nom}}| EL_{\text{spec}}} + 1} \quad \text{and} \quad CD_{\text{rel-iso-limit}} = \frac{|CD_1 - CD_{\text{limit}}|}{CD_{\text{nom}}} \quad (8.52)$$

and CD_{limit} is the CD value of the process window contour closest to the isofocal CD. This K factor shrinks the process window as a function of how far CD_1 is from its closest CD specification limit, and thus how much the process window is bent upward or downward.

8.5.4 Overlapping Process Windows

Although all of the above results describe the focus and exposure response of one critical feature, in reality a number of mask features must be printed simultaneously. Using the most common example, isolated lines typically occur on the same design as dense lines. Thus, a more important measure of performance than the DOF of each individual feature is the *overlapping* DOF of multiple critical features. Just as multiple profile metrics were overlapped to form one overlapping process window in Figure 8.20b, process windows from different features can be overlapped to determine the DOF for simultaneously printing those multiple features. Figure 8.26 shows such an example for isolated and dense features, where subresolution assist features (SRAFs) are used to reduce the isofocal bias of the isolated line and achieve a significantly better overlapping DOF. Ideally, process windows for all critical feature sizes and pitches would be overlapped to find out which of the features represent the process window limiters.

Systematic variations in patterning as a function of field position can also be accounted for with overlapping process windows. If the same feature were printed at different points in the field (typically the center and the four corners of the field are sufficient), process

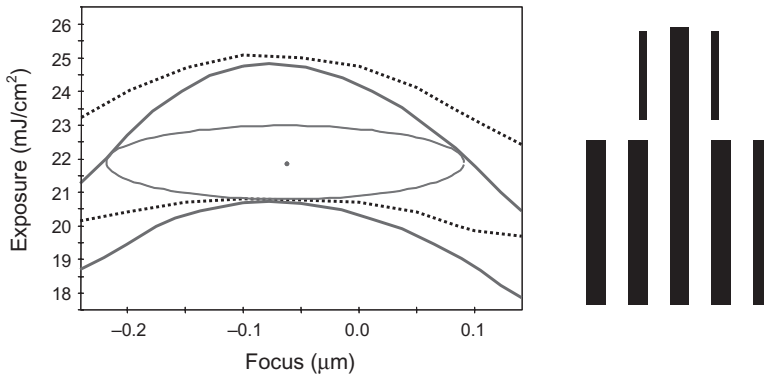


Figure 8.26 The overlapping process window for the dense (dashed lines) and isolated (solid lines) features shown to the right of the graph

windows can be overlapped to create the *usable* process window for that feature. The resulting depth of focus is called the *useable depth of focus* (UDOF).

8.5.5 Dose and Focus Control

Controlling focus and dose is an important part of keeping the critical dimensions of the printed patterns in control. The first step in understanding how to control focus and dose is to characterize the response of critical features to variations using the focus–exposure process window, as discussed above. Proper analysis of CD versus focus and dose data allows for the calculation of the process window, measurement of the process window size to produce the exposure latitude–DOF plot, and determination of a single value for the depth of focus. Best focus and dose are also determined with this analysis as the process settings that maximize the tolerance to focus and dose errors. Once best focus and dose are determined, the next goal is to keep the process centered at this best focus and dose condition as thousands of production wafers pass through the lithography cell. And since almost all errors that occur in lithography act like either a dose error or a focus error, properly adjusting dose and focus on a lot-by-lot basis can provide for much tighter CD control. Dose and focus monitoring and feedback control has become a quite common technique for living within shrinking process windows.

The most common way to monitor and correct for drifts in exposure dose grew out of standard product-monitoring metrology applications. Critical dimension test structures (generically called *metrology targets*) are placed in the kerf (streets separating each die) and are designed to mimic the size and surroundings of some critical feature of interest within the die. For example, on the polysilicon level of a CMOS device, the most common critical gate may be a 90-nm line on a 250-nm pitch. The CD targets might then be an array of five 90-nm lines on a 250-nm pitch. These types of targets are called *device-representing*.

By characterizing the change in CD versus dose of this specific device-representing target, any change in measured CD can be directly related to an equivalent change in dose (see Figure 8.3). Often the data are fit to a function, and the function used to predict the

required dose correction. While a linear function is often used (everything becomes linear over a small enough range), a significantly better function was given in Equation (8.31). This equation can be inverted to give the dose correctable ΔE for a measured critical dimension error ΔCD .

$$\Delta E = E_1 \left(\frac{\Delta CD}{s CD_1 - \Delta CD} \right) \quad \text{where} \quad s = \left. \frac{d \ln CD}{d \ln E} \right|_{E=E_1} \quad (8.53)$$

Multiple measurements and a proper sampling plan help to eliminate random errors in the measured CD and ensure that the changes observed are systematic. By monitoring this target on product lots over time, temporal drifts in CD can be corrected by changing the exposure dose in a feedback loop (often automated using *advanced process control*, APC).

The advantages of using device-representing targets are their simplicity and multipurpose use. Device-representing CD targets are often measured for the purpose of lot monitoring: do the measured CD values indicate that this lot will yield high or low? If these structures are being measured anyway, it only makes sense to use them for dose monitoring and corrections as well. The disadvantage of using device-representing targets, as we shall see below, is that they may have reduced sensitivity to the variable being controlled (exposure dose, in this case), compared to other possible test structures, and may be too sensitive to other variables not being measured or controlled (focus being the most important of these).

Monitoring exposure dose is relatively easy since all features vary monotonically with dose. However, as we saw above, the sensitivity of any given feature to dose depends on focus. In other words, the exposure latitude slope term in Equation (8.53) is focus dependent. In general, s varies quadratically with focus, reaching a minimum at best focus. Further, focus errors most often cause CD errors. If focus drifts, the use of an exposure monitoring strategy like that outlined above will make exposure dose adjustments to compensate for focus errors. If sufficiently large, the combined focus drift and exposure adjustment can render the lithography process unstable (meaning the useable process window becomes too small to ensure adequate CD control) even though the target CD appears to be in control. How can this situation be avoided?

As we saw above, the two variables focus and dose must be characterized together in a focus–exposure matrix (Figure 8.17). Note that at some doses, changes in focus make this CD larger, while at other doses the same change in focus will make the CD smaller. At the isofocal dose, changes in focus have the least impact on CD. This observation suggests a way to improve exposure monitoring: If a CD target is found which exhibits isofocal behavior at the nominal dose, this test structure will have minimum sensitivity to focus errors while maintaining reasonable dose sensitivity. Such a test structure will be called an *isofocal target*. An isofocal target will have minimum sensitivity to focus errors, so that only dose errors will be detected and corrected. However, focus errors will still change the exposure latitude term s in the exposure correction equation. Thus, if both focus and exposure errors are present, the predicted dose correction from Equation (8.53) will be too large. To address this deficiency, focus monitoring must be added to the overall control strategy.

Focus presents an extremely difficult control problem since CD tends to vary quadratically with focus. As a result, it becomes extremely difficult to determine the *direction* of a focus error – was the focal plane too high or too low compared to best focus? Thus, the focus monitoring problem is often broken up into two parts: determining the magnitude of a focus error, and its direction.

Since the responses of CD to dose and to focus are coupled, the goal is to find two targets with different responses to these variables. If, for example, two targets had the same response to dose, but different responses to focus, measuring them both would allow focus errors to be separated from dose errors. A simple approach would be to find a dense line whose isofocal dose matched the nominal dose for the process and an isolated line far removed from its isofocal dose. Further, if the isolated line were sized so that it had approximately the same exposure latitude as the dense line [same value of s in Equation (8.53)], then the isofocal dense feature could be used as the dose monitor while the difference in CD between dense and isolated features could be used to monitor focus. Figure 8.27 shows an example of a reasonably well-designed target. The dense feature (which is the dose monitor) varies by only about 2nm over the full range of focus errors of interest, but varies by over 20nm over the exposure range of interest. The CD difference (the focus monitor) varies by only 2 nm over the dose range of interest but changes by over 20nm over the focus range of interest. Thus each monitor has 10× *discrimination*, that is, 10 times more sensitivity to the variable being monitored compared to the other variable. Further refinement of the target design can often lead to even better discrimination between focus and dose errors, while maintaining high sensitivity.

It is possible to improve the analysis of this dual-target dose and focus monitoring strategy by creating models for the combined focus–exposure response of each target. Inverting both models using the two CD target measurements can provide for good discrimination between dose and focus errors even when the two features do not have identical exposure latitude and the dense feature is not perfectly isofocal. It is important to remember, however, that the nearly quadratic response of CD with focus means that focus direction cannot be determined from such data.

To monitor the direction of a focus error, one must measure a response on the wafer that is sufficiently asymmetric with respect to focus direction. CD does not provide such a response in general. Of course, at any time the lithography engineer can stop running product wafers and print a focus–exposure matrix wafer in order to determine best focus and adjust the focus on the stepper or scanner. However, the goal of product monitoring is to make measurements on product wafers that can detect and correct for dose and focus errors quickly without reducing stepper productivity or throughput. Some novel approaches have been developed, each with their own advantages and disadvantages. Two important examples are described below.

The phase-shift focus monitor developed by Tim Brunner¹⁰ uses a 90° phase shift pattern on the mask to turn focus errors into pattern placement errors on the wafer. Thus, a measurement of overlay for this pattern is directly proportional to the focus error, direction included. Unfortunately, the need for a 90° phase shift region on the mask means that this focus monitor cannot in general be placed on product reticles – it is a test mask approach that doesn't meet product monitoring requirements.

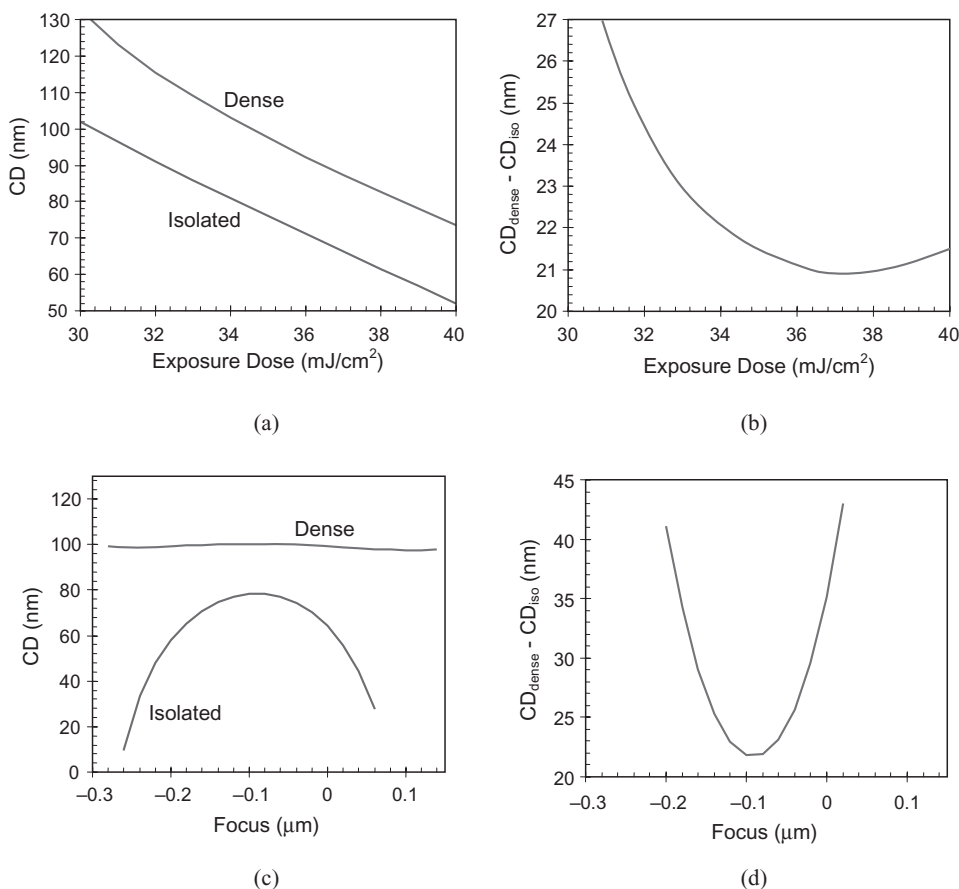


Figure 8.27 A dual-target approach to monitoring dose and focus using a 90-nm dense line on a 220-nm pitch and an 80-nm isolated line: (a) dense and isolated lines through dose; (b) the iso-dense difference through dose; (c) dense and isolated lines through focus; (d) the iso-dense difference through focus

Scatterometry-based measurements offer a product-measurement solution to the focus monitoring problem. Unlike scanning electron microscopes, scatterometry allows for the simultaneous measurement of linewidth, sidewall angle and resist height for a pattern of long lines and spaces. While CD varies approximately quadratically with focus, resist height and sidewall angle have a strong linear component to their responses, making detection of focus direction possible. Like Figure 8.19 above, Figure 8.28 shows an example of how a typical resist cross section varies with focus. The strong asymmetry of the sidewall angle can provide sensitive detection of focus direction.

The sensitivity of a monitor target is defined as the smallest error that can be reliably detected. The goal of focus and exposure monitoring is to have sensitivities of detecting focus and exposure errors equal to about 10–20% of the process window size or less. For example, if the depth of focus is 400 nm, the focus sensitivity must be at least 80 nm, but

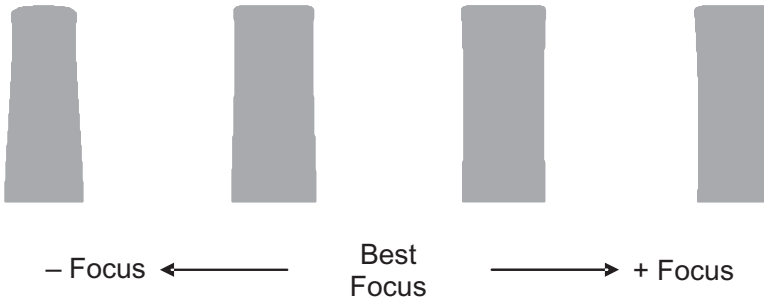


Figure 8.28 The asymmetric response of resist sidewall angle to focus provides a means for monitoring focus direction as well as magnitude. Here, + focus is defined as placing the focal plane above the wafer (moving the wafer further away from the lens)

40nm is preferable. For an 8% exposure latitude specification, the dose sensitivity must be at least 1.6%, but 0.8% is the preferred sensitivity.

8.6 H–V Bias

One interesting effect of aberrations is called horizontal–vertical (H–V) bias. Quite simply, H–V bias is the systematic difference in linewidth between closely located horizontally and vertically oriented resist features that, other than orientation, should be identical. There are two main causes of H–V bias: astigmatism and related aberrations; and source shape aberrations.

8.6.1 Astigmatism and H–V Bias

The aberration of astigmatism results in a difference in best focus as a function of the orientation of the feature. Using the Zernike polynomial description of aberrations (see Chapter 3), 3rd order 90° astigmatism (which affects horizontally and vertically oriented lines) takes the form

$$\text{phase error} = 2\pi Z_4 R^2 \cos 2\phi \quad (8.54)$$

Consider a vertically, y -oriented pattern of lines and spaces. The diffraction pattern will spread across the x -axis of the pupil, corresponding to $\phi = 0^\circ$ and 180° . Thus, the phase error will be $2\pi Z_4 R^2$ for this feature. Recalling the description of defocus as an aberration (see Chapter 3), the phase error due to defocus is

$$\text{phase error} \approx \frac{\pi \delta NA^2}{\lambda} R^2 \quad (8.55)$$

where δ is the defocus distance, λ is the wavelength, and NA is the numerical aperture. [Equation (8.55) is approximate because it retains only the first term in a Taylor series. While this approximation is progressively less accurate for higher numerical apertures, it will be good enough for our purposes.] Immediately, one sees that 3rd order astigmatism

looks just like the approximate effect of defocus. Thus, astigmatism will cause the vertically oriented lines to shift best focus by an amount

$$\Delta\delta_{\text{vert}} \approx \frac{2Z_4\lambda}{NA^2} \quad (8.56)$$

For horizontally oriented lines, the diffraction pattern will be along the y-axis of the pupil ($\theta = \pm 90^\circ$) and the astigmatism will cause a phase error of $-2\pi Z_4 R^2$. Thus, the focus shift for the horizontal lines will be the same magnitude as given by Equation (8.56), but in the opposite direction.

To see how astigmatism causes H-V bias, we need to understand how a shift in focus might affect the resist feature CD. To first order, CD has a quadratic dependence on focus.

$$CD \approx CD_{\text{best focus}} + a\delta^2 \quad (8.57)$$

where a is the dose-dependent curvature of the CD through focus curve (see section 8.5.1) and is given by

$$a = CD_1 \left(\frac{d \ln CD}{d \ln E} \bigg|_{E_1, F_0} \right) \left(1 - \frac{E_1}{E} \right) \left(\frac{1}{\Delta^2} \right) \quad (8.58)$$

Recalling the typical shapes of Bossung curves, a can vary from positive to negative values as a function of dose (that is, depending on whether E is greater than or less than the isofocal dose, E_1). If best focus is shifted due to astigmatism, we can calculate the CD of the vertical and horizontal features by adding the focus shift of Equation (8.56) to Equation (8.57).

$$\begin{aligned} CD_{\text{vert}} &\approx CD_{\text{best focus}} + a \left(\delta - \frac{2Z_4\lambda}{NA^2} \right)^2 \\ CD_{\text{horiz}} &\approx CD_{\text{best focus}} + a \left(\delta + \frac{2Z_4\lambda}{NA^2} \right)^2 \end{aligned} \quad (8.59)$$

From, here a straightforward subtraction gives us the H-V bias:

$$H - V \text{ bias} \approx \frac{8a\delta Z_4\lambda}{NA^2} \quad (8.60)$$

The H-V bias is directly proportional to the amount of astigmatism in the lens (Z_4) and to the curvature of the CD-through-focus curve (a). But it is also directly proportional to the amount of defocus. In fact, a plot of H-V bias through focus is a sure way to identify astigmatism (that is, when not using the isofocal dose, where $a \approx 0$, for the experiment). Figure 8.29 shows some typical results. Note that the isolated lines show a steeper slope than the dense lines due to a larger value of the CD through focus curvature. In fact, if a is determined by fitting Equation (8.57) to CD through focus data, a reasonable estimate of Z_4 can be made using an experimentally measured H-V bias through focus curve. Note also that the true shape of the curves in Figure 8.29 is only approximately linear, since both Equations (8.56) and (8.57) ignore higher-order terms.

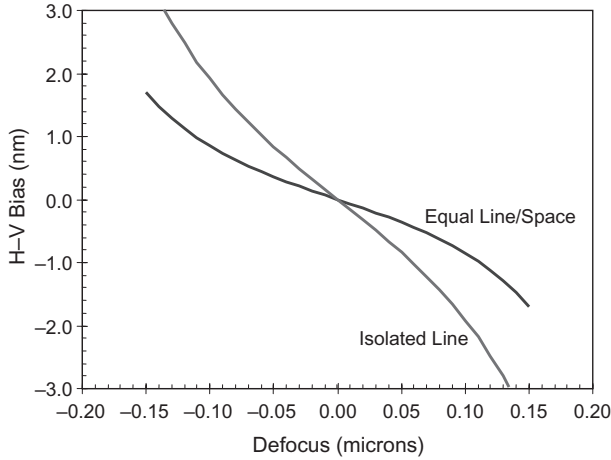


Figure 8.29 PROLITH simulations of H–V bias through focus showing approximately linear behavior ($\lambda = 193\text{ nm}$, $NA = 0.75$, $\sigma = 0.6$, 150-nm binary features, 20 milliwaves of astigmatism). Simulations of CD through focus and fits to Equation (8.57) gave the CD curvature parameter $a = -184\text{ }\mu\text{m}^{-2}$ for the dense features and $-403\text{ }\mu\text{m}^{-2}$ for the isolated lines

Assume that the maximum possible value of the defocus is one-half of the depth of focus (DOF). At this defocus, Equation (8.57) would tell us that at the worst case dose, the term $a\delta^2$ will be about 10% of the nominal CD (by definition of the process window with a $\pm 10\%$ CD specification). Thus, we can say that within the process window, the worst case H–V bias due to astigmatism will be

$$\left. \frac{H-V \text{ bias}}{CD_{\text{nominal}}} \right|_{\text{max}} \approx \frac{1.6Z_4\lambda}{DOF NA^2} \quad (8.61)$$

Using typical numbers for a 65-nm process, for a wavelength of 193 nm, an NA of 0.9, and assuming a depth of focus of 200 nm, the fractional H–V bias will be about $2Z_4$. If we are willing to give 1% CD error to H–V bias, our astigmatism must be kept below 5 milliwaves. In general, Equation (8.61) shows that as new, higher resolution scanners are designed and built, the astigmatism in the lens must shrink as fast or faster than the depth of focus of the smallest features to be put into production.

8.6.2 Source Shape Asymmetry

The second major cause of H–V bias is illumination aberrations (that is, source shape asymmetries). In Chapter 2, a discussion of partial coherence showed that source points must be symmetrically placed about the axis of symmetry for a mask pattern in order to avoid telecentricity errors (which cause pattern placement errors as a function of defocus). For example, for y-oriented lines and spaces, a source point at (f_x, f_y) must be accompanied by a corresponding source point of the same intensity at $(-f_x, f_y)$ in order to avoid telecentricity errors. The avoidance of telecentricity errors for x-oriented lines will likewise

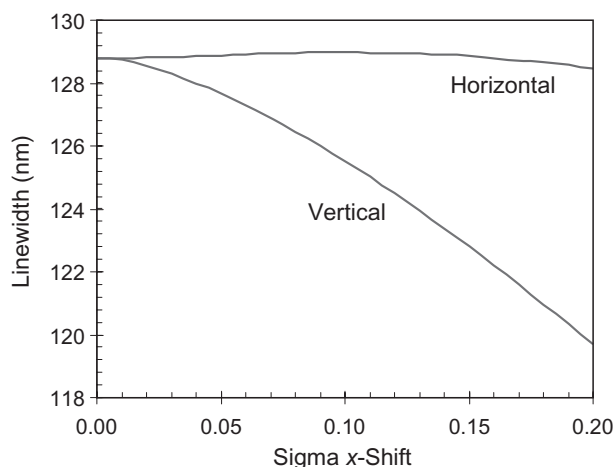


Figure 8.30 Example of how an x -shift in the center of a conventional source ($\sigma = 0.6$) affects mainly the vertical (y -oriented) lines and spaces ($CD = 130\text{ nm}$, $\text{pitch} = 650\text{ nm}$, PROLITH simulations)

require additional points at $(f_x, -f_y)$ and $(-f_x, -f_y)$, providing the general guidance that all source shapes must have fourfold symmetry. If both horizontal and vertical lines must be printed simultaneously, identical printing will occur only if the source shape is invariant to a 90° rotation. This provides an additional constraint that for every source point (f_x, f_y) , there must be a corresponding source point of the same intensity at $(-f_y, f_x)$.

Consider conventional illumination where the center of the disk-shaped source is not perfectly aligned with the center of the optical path (called an illumination telecentricity error). The impact of such a source telecentricity error is dependent on the partial coherence, the pitch, and of course on the amount of telecentricity error. Figure 8.30 shows that, in general, an x -shift of the center of the illumination source shape affects vertically oriented (y -oriented) lines significantly, but horizontal (x -oriented) lines very little.

For the simple case of dense line/space patterns where only the zero and the two first diffraction orders are used in the imaging, the image will be made up of combinations of one-, two- and/or three-beam interference. The change in CD for the vertical features for an x -shift in the source center is caused by a change in the ratio of two-beam to three-beam imaging. Figure 8.31 shows an example case where the total three-beam imaging area (light grey) and the total two-beam imaging area (dark grey) do not appreciably change as the center of the source is shifted by about 0.1 sigma. Thus, for this pitch and sigma, one would not expect to see much change in the vertical feature CD. Consider a different pitch, as shown below in Figure 8.32a. At this particular pitch, all of the first order is inside the lens, so that all of the imaging is three-beam (note that the second diffracted order is not shown in the diagram for clarity's sake). However, when the source is shifted in x by 0.1 sigma (Figure 8.32b), the amount of three-beam imaging for the vertical lines is reduced and two-beam imaging is introduced. By contrast, the horizontal lines (which spread the diffraction pattern vertically in the pupil) have only an

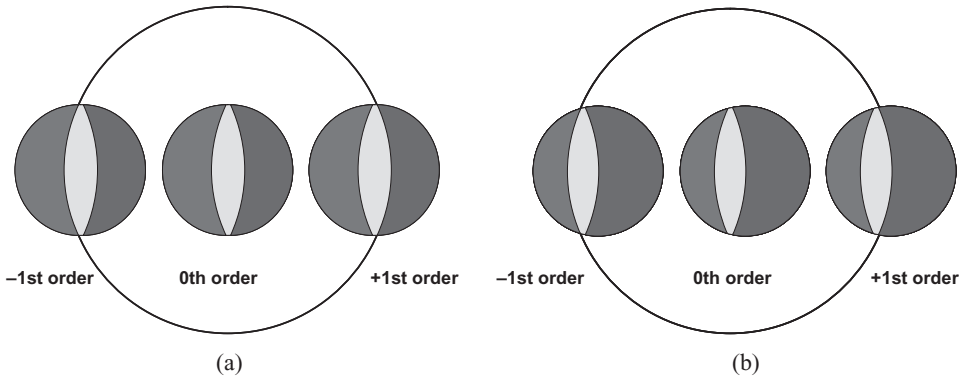


Figure 8.31 Example of dense line/space imaging where only the zero and first diffraction orders are used ($k_{pitch} = 1.05$). The middle segment of each source circle represents three-beam imaging, the outer areas are two-beam imaging. (a) source shape is properly centered, (b) source is offset in x (to the right) by 0.1. Note that the diffraction pattern represents vertical (y -oriented) features

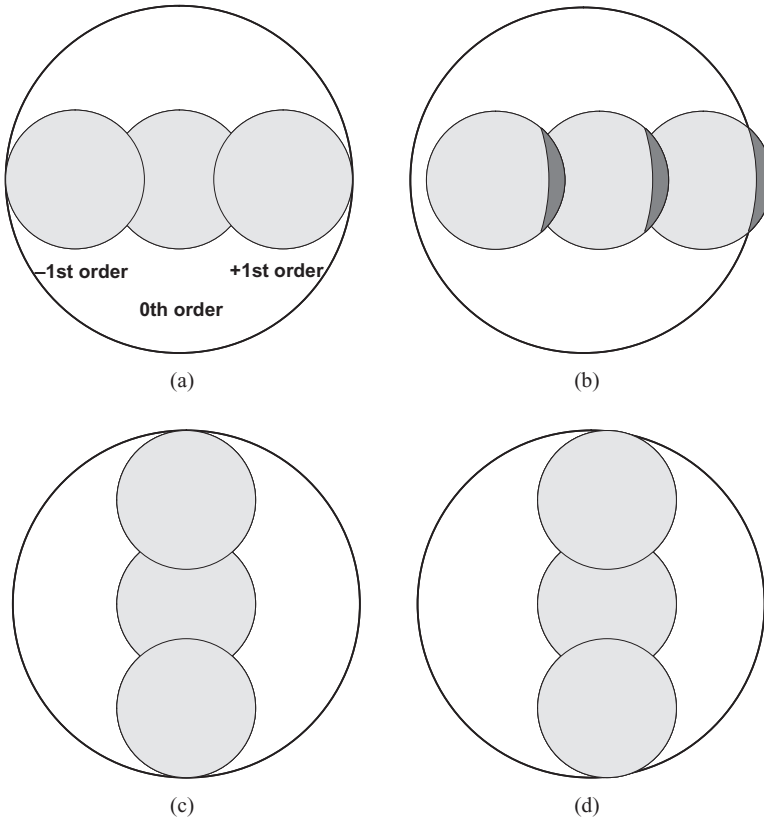


Figure 8.32 Examples of how a telecentricity error affects the ratio of two-beam to three-beam imaging at the worst case pitch ($\sigma = 0.4$, $x\text{-shift} = 0.1$, $k_{pitch} = 1.65$). (a) vertical lines, no telecentricity error, (b) vertical lines, with telecentricity error, (c) horizontal lines, no telecentricity error, and (d) horizontal lines, with telecentricity error

imperceptible change in the amount of three-beam imaging for the same x -shift of the source position (Figure 8.32c and d). Thus, at this pitch, one would expect to see a large amount of H–V bias with source shift.

The pitch that just allows only three-beam imaging for a given partial coherence is the pitch that is most sensitive to a shift in the source center position. Thus, the worst case pitch from an H–V bias telecentricity error sensitivity perspective is given by

$$\text{worst case } k_{\text{pitch}} = \frac{pNA}{\lambda} \approx \frac{1}{1-\sigma} \quad (8.62)$$

8.7 Mask Error Enhancement Factor (MEEF)

Errors in the critical dimensions of the features on a mask have always been a significant cause of critical dimension errors on the wafer. As optical lithography has pushed features to lower and lower k_1 values (= feature size $\cdot NA/\lambda$), however, mask CD errors have taken up a much larger fraction of the overall sources of CD variations on the wafer.

8.7.1 Linearity

An important constraint placed on any lithographic imaging task is that *every* feature on the mask must be faithfully imaged onto the wafer within predetermined tolerances. Most typical integrated circuit device layers consist of a myriad of different pattern types, shapes and sizes. It is a common assumption that since the resolution defines the smallest pattern that can be acceptably imaged, all features larger than this limit will also be acceptably imaged. Unfortunately, this assumption may not always be true. For imaging systems designed to maximize the printability of a given small feature (e.g. using off-axis illumination), often some larger features will be more difficult to print. One way to insure that larger features print well at the same time as the minimum feature is to build this requirement into the definition of resolution: *the smallest feature of a given type such that it and all larger features of the same type can be printed simultaneously with a specified depth of focus*. This resolution is called the *linear resolution* of the imaging system.

The linear resolution is typically assessed using the mask linearity plot. Consider a mask with many feature sizes of a given type, for example, equal lines and spaces, isolated lines, or contact holes. By plotting the resulting resist feature width versus the mask width, the mask linearity plot is generated. Shown in Figure 8.33 are examples of linearity plots for equal line/space patterns and isolated lines, both imaged at best focus and at the dose-to-size for the 350-nm features. Perfect, linear behavior would be a line through the origin with a slope of 1. By defining specifications for any deviation from perfect linearity ($\pm 5\%$, for example), the minimum feature that stays within the specification would be the linear resolution. Qualitatively, Figure 8.33 shows that the equal line/space patterns of this example have a linear resolution down to about 350 nm ($k_1 = 0.54$), whereas the isolated lines are linear down to about 300 nm ($k_1 = 0.46$).

Of course, to be truly practical, one should include variations in exposure and focus as well. The most rigorous approach would begin with the focus–exposure process window

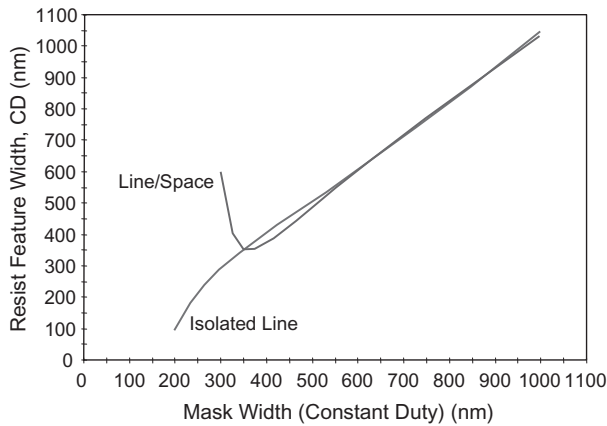


Figure 8.33 Typical mask linearity plot for isolated lines and equal lines and spaces (*i*-line, $NA = 0.56$, $\sigma = 0.5$)

for the largest feature size. The process window from each smaller feature is then overlapped with that from the larger features and the depth of focus calculated. Smaller and smaller features are added until the overlapped depth of focus drops below the specified limit, indicating the linear resolution limit. Thus, although mask linearity plots do not provide a rigorous, general method for determining the linear resolution, they are qualitatively useful.

As lithography for manufacturing continues to push toward its ultimate resolution limits, linearity is playing a decidedly different role in defining the capabilities of low k_1 imaging. Consider, using Figure 8.33 as an example, manufacturing at the linear resolution limits: 350-nm lines and spaces and 300-nm isolated lines. Although these features may be ‘resolvable’ by the definitions provided above, critical dimension (CD) control may be limited by a new factor: how do errors in the dimensions of the feature on the mask translate into errors in resist CD on the wafer?

8.7.2 Defining MEEF

For ‘linear’ imaging, mask CD errors would translate directly into wafer CD errors (taking into account the reduction factor of the imaging tool, of course). Thus, a 3-nm CD error on the mask (all CDs on the mask will be expressed here in wafer dimensions) would result in a 3-nm CD error on the final resist feature. If, however, the features of interest are at the very edge of the linear resolution limit, or even beyond it, the assumption of linear imaging falls apart. How then do mask CD errors translate into resist CD errors?

Consider the examples shown in Figure 8.33. If an isolated line is being imaged near its resolution limit, about 300 nm, a 10-nm mask CD error would give a 14-nm resist CD error. Thus, at this feature width, isolated line mask errors are amplified by a factor of 1.4. This amplification of mask errors is called the *mask error enhancement factor*

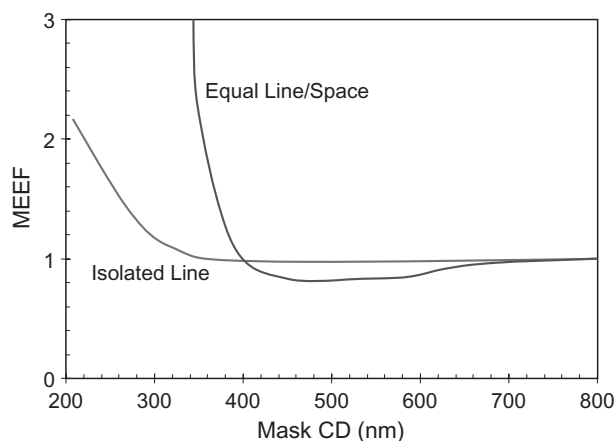


Figure 8.34 The mask error enhancement factor (MEEF) for the data of Figure 8.33

(MEEF). First discussed by Wilhelm Maurer,¹¹ the MEEF (also called MEF by some authors) is defined as the change in resist CD per unit change in mask CD:

$$MEEF = \frac{\partial CD_{\text{resist}}}{\partial CD_{\text{mask}}} \quad (8.63)$$

where again the mask CD is in wafer dimensions. If we assume a ‘zero bias’ process (where the target CD on the wafer is the same as the mask CD), then the MEEF can also be expressed as

$$MEEF = \frac{\partial \ln CD_{\text{resist}}}{\partial \ln CD_{\text{mask}}} \quad (8.64)$$

This form of the MEEF equation will be useful later. Figure 8.34 shows how the MEEF varies with feature size for the mask linearity data of Figure 8.33. Regions where the MEEF is significantly greater than 1 are regions where mask errors may come to dominate CD control on the wafer. Optical proximity correction techniques (discussed in Chapter 10) allow us to lower the linear resolution, but generally without improving the MEEF. As a result, the mask has begun to take on a much larger portion of the total CD error budget.

8.7.3 Aerial Image MEEF

A MEEF of 1.0 is the definition of a linear imaging result. Although a MEEF less than one can have some desirable consequences for specific features, in general a MEEF of 1.0 is best. Fundamentally, anything that causes the overall imaging process to be nonlinear will lead to a nonunit valued MEEF. In lithography, every aspect of the imaging process is nonlinear to some degree, with the degree of nonlinearity increasing as the dimensions of the features approach the resolution limits. Consider the first step in the imaging process, the formation of an aerial image. One might judge the linearity of this

first step by approximating the resist CD with an image CD, defined to be the width of the aerial image at some image threshold intensity value. It is important to note that the image CD is only an approximate indicator of the resist CD. For real, finite-contrast resists the differences between these two quantities can be substantial. Nonetheless, the image CD will be used here to elucidate some general principles about imaging and the MEEF.

For two simple cases of projection imaging, coherent and incoherent illumination, analytical expressions for the aerial image can be defined. Assuming a pattern of many long lines and spaces with a spacewidth w and pitch p such that only the 0 and ± 1 diffraction orders pass through the lens, the TE coherent and incoherent in-focus aerial images would be

$$\text{Coherent Illumination: } I(x) = \left[\frac{w}{p} + \frac{2 \sin(\pi w/p)}{\pi} \cos(2\pi x/p) \right]^2 \quad (8.65)$$

$$\text{Incoherent Illumination: } I(x) = \frac{w}{p} + \frac{2 \sin(\pi w/p)}{\pi} (MTF_1) \cos(2\pi x/p) \quad (8.66)$$

where MTF_1 is the value of the incoherent Modulation Transfer Function at the spatial frequency corresponding to the first diffraction order. The requirement that no orders higher than the first diffraction order be used to form the image means that the coherent image equation is valid for a limited range of pitches such that $1 < pNA/\lambda < 2$, and the incoherent expression is valid for $0.5 < pNA/\lambda < 1$.

Using these expressions to define the image CD, exact expressions for the *image MEEF* can be derived for these repeating line/space patterns under the conditions given above:

$$\text{image MEEF} = \frac{\partial CD_{\text{image}}}{\partial CD_{\text{mask}}} = \frac{\partial CD_{\text{image}}}{\partial w} \quad (8.67)$$

$$\text{Coherent Illumination: image MEEF} = \frac{2 + \cos(2\pi w/p)}{1 - \cos(2\pi w/p)} \quad (8.68)$$

$$\text{Incoherent Illumination: image MEEF} = \frac{\frac{1}{MTF_1} + 1 + \cos(2\pi w/p)}{1 - \cos(2\pi w/p)} \quad (8.69)$$

An interesting observation can be made immediately. Over the range of valid pitches, the coherent image MEEF is only dependent upon the duty cycle (w/p), not on the pitch itself. The incoherent image MEEF, on the other hand, has a direct pitch dependence through the value of the MTF (which is approximately equal to $1 - \lambda/[2NAp]$). Figure 8.35 shows how both image MEEFs vary with spacewidth-to-linewidth ratio.

The extreme nonlinearity of the imaging process is evident from the results shown in Figure 8.35. For coherent illumination, a pattern of equal lines and spaces will have an image MEEF of 0.5. A spacewidth twice the linewidth produces a MEEF of 1.0, and a spacewidth three times the linewidth results in a coherent image MEEF of 2.0. Obviously, different duty cycles can have wildly different sensitivities to mask errors. While the

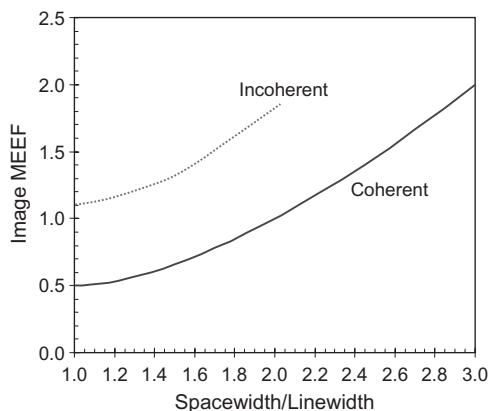


Figure 8.35 The impact of duty cycle (represented here as the ratio of spacewidth to linewidth for an array of line/space patterns) on the image CD-based MEEF for both coherent and incoherent illumination. For the incoherent case, an MTF_1 of 0.45 was used

approximations used do not apply to truly isolated lines, it is clear that such features will also deviate from unit MEEF. A spacewidth/linewidth ratio less than unity also enhances the effect.

Although neither purely coherent nor purely incoherent illumination is ever used in real lithographic imaging, these two extremes tend to bound the behavior of typical partially coherent imaging tools. Thus, we would expect the image MEEF of a partially coherent imaging system to vary with both duty cycle and pitch, and to vary by about a factor of 2 as a function of the partial coherence.

8.7.4 Contact Hole MEEF

Possibly the most challenging mask layer to print in high-end lithography processes is the contact or via layer. These small features are subwavelength in two dimensions, making them exceptionally sensitive to everything that makes low k_1 lithography difficult. For example, contact holes suffer from the largest MEEFs of any feature type (as we shall see below). Assuming that the source of the mask error is isotropic (that is, it affects both the height and width of the contact in the same way), then such an error is affecting two dimensions of the mask simultaneously. In other words, errors in the area of the mask feature go as errors in the CD squared.

When printing a small contact hole, the aerial image projected onto the wafer is essentially the same as the point spread function (PSF) of the optical system. The point spread function is defined as the normalized aerial image of an infinitely small contact hole on the mask. For small contact holes (about $0.6\lambda/NA$ or smaller), the aerial image takes on the shape of the PSF. If the contact hole size on the mask is smaller than this value, the printed image is controlled by the PSF, not by the dimensions of the mask. Making the contact size on the mask smaller only reduces the intensity of the image peak. This results in a very interesting relationship: a change in the mask size of a small contact hole is essentially equivalent to a change in exposure dose.

It is the area of the contact hole that controls the printing of the contact. The electric field amplitude transmittance of a small contact hole is proportional to the area of the contact, which is the CD squared. The intensity is then proportional to the area squared. Thus, the effect of a small change in the mask CD of a contact hole is to change the effective dose reaching the wafer (E) as the CD to the fourth power.

$$\begin{aligned} E &\propto CD_{\text{mask}}^4 \\ d \ln E &\approx 4 d \ln CD_{\text{mask}} \end{aligned} \quad (8.70)$$

where the approximate sign is used because the fourth power dependence is only approximately true for typically sized contact holes.

The above equation relates mask errors to dose errors. Combining this with the MEEF Equation (8.64) allows us to relate MEEF to exposure latitude.

$$MEEF \approx 4 \frac{\partial \ln CD_{\text{wafer}}}{\partial \ln E} \quad (8.71)$$

The term $\partial \ln CD / \partial \ln E$ can be thought of as the percent change in CD for a 1 % dose change and is the inverse of the common exposure latitude metric. Thus, anything that improves the exposure latitude of a contact hole will also reduce its MEEF. Anything that reduces exposure latitude (like going out of focus) will result in a proportional increase in the MEEF. The importance of exposure latitude as a metric of printability is even greater for contact holes due to the factor of four multiplier for the MEEF. Additionally, the above expression could be used to lump mask errors into effective exposure dose errors for the purpose of process window specifications. Mask errors could be thought of as consuming a portion of the exposure dose budget for a process.

8.7.5 Mask Errors as Effective Dose Errors

The concept of mask errors as effectively exposure dose errors, first discussed above in the context of contact hole MEEF, can be used as a general concept for understanding the nature of the mask error enhancement factor. Consider again the image MEEF based on an aerial image CD. For a given mask feature size w , the image CD will occur at $x = CD/2$ when the intensity equals the image threshold value.

$$I(x = CD/2, w) = I_{\text{th}} \quad (8.72)$$

Consider the total differential of the intensity with x and w as variables:¹²

$$dI = \frac{\partial I}{\partial x} dx + \frac{\partial I}{\partial w} dw = 0 \quad (8.73)$$

where the total differential must be zero since the intensity is constrained to be at the threshold intensity. Thus,

$$\frac{dx}{dw} = - \frac{\partial I / \partial w}{\partial I / \partial x} \quad \text{or} \quad \text{image MEEF} = -2 \frac{\partial I / \partial w}{\partial I / \partial x} \quad (8.74)$$

In this expression, the numerator can be thought of as the change in dose reaching the wafer caused by a change in the mask CD (Figure 8.36). The denominator is the image

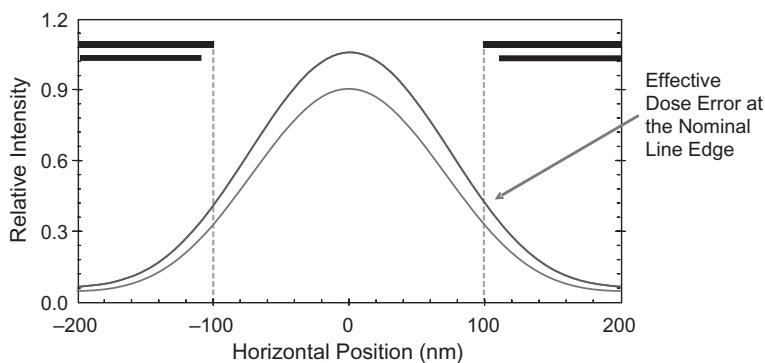


Figure 8.36 Mask errors can be thought of as creating effective dose errors near the edge of the feature

slope, which is proportional to exposure latitude. An alternate form of this expression is

$$\text{image MEEF} = \frac{2}{NILS} \left. \frac{\partial \ln I}{\partial \ln w} \right|_{x=CD/2} \quad (8.75)$$

As an example, consider the out-of-focus coherent three-beam image for TE illumination:

$$I(x) = \left[\frac{w}{p} + \frac{2 \sin(\pi w/p)}{\pi} \cos(\pi \lambda \delta / p^2) \cos(2\pi x/p) \right]^2 \quad (8.76)$$

(The low NA expression for defocus is used here, but the results will be identical if the more exact expression were used.) Taking the derivative with respect to the mask spacewidth w ,

$$\left. \frac{\partial \ln I}{\partial \ln w} \right|_{x=w/2} = \frac{2w}{p} \frac{[1 + 2 \cos^2(\pi w/p) \cos(\pi \lambda \delta / p^2)]}{\sqrt{I(w/2)}} \quad (8.77)$$

For the case of equal lines and spaces, this derivative becomes a constant (since the edge of the feature is at the image isofocal point):

$$\left. \frac{\partial \ln I}{\partial \ln w} \right|_{x=CD/2} = 2 \quad (8.78)$$

Thus, the coherent three-beam equal line/space image MEEF, including defocus, becomes

$$\text{image MEEF} = \frac{4}{NILS} \quad (8.79)$$

As will be explained in greater detail in Chapter 9, $2/NILS$ is equal to the slope of a log-CD-versus-log-dose curve for the case of an infinite contrast resist. Thus, one can

generalize this image MEEF by replacing $2/NILS$ with the actual CD-versus-dose slope.

$$MEEF \approx \left(\frac{\partial \ln CD_{\text{wafer}}}{\partial \ln E} \right) \left(\frac{\partial \ln I}{\partial \ln w} \right)_{x=CD/2} \quad (8.80)$$

For the case of a small contact hole, we saw in the previous section that the fractional dose change per fractional mask CD change was 4. For a small isolated space, the value is about 2. For other features, the change in intensity at the nominal line edge can be computed and an estimate of MEEF made using Equation (8.80).

Another way to interpret the above result is to consider that mask errors result in a reduction of the effective exposure latitude. If EL is the exposure latitude when no mask errors are present, then the effective exposure latitude in the presence of mask errors of range ΔCD_{mask} will be

$$EL_{\text{eff}} \approx EL \left(1 - MEEF \frac{\Delta CD_{\text{mask}}}{\Delta CD_{\text{spec}}} \right) \quad (8.81)$$

where ΔCD_{spec} is the CD tolerance used to define the exposure latitude (for example, 0.2 times the nominal CD). This expression can be used to scale an entire exposure latitude versus depth of focus plot, such as Figure 8.22, and thus can also be used to determine the effect of mask errors on the available depth of focus.

8.7.6 Resist Impact on MEEF

For a given aerial image response, the response of a real, finite-contrast resist will change the value of the MEEF, sometimes dramatically. Although it is difficult to systematically and controllably vary resist contrast experimentally, simulation provides an effective tool for exploring its impact on MEEF theoretically. Consider a simple baseline process of 248-nm exposure with a 0.6 NA, 0.5 partial coherence imaging tool using 500-nm-thick resist on BARC on silicon. The contrast of the resist is controlled by essentially one simulation parameter, the dissolution selectivity parameter n of the original kinetic dissolution rate model (see Chapter 7). High values of n correspond to high values of resist contrast. For this example, $n = 5$ will be a low-contrast resist, $n = 10$ is a mid-contrast resist and $n = 25$ will be called a high-contrast resist.

The variation of MEEF with nominal mask feature size for these three virtual resists is shown in Figure 8.37 for a mask pattern of equal lines and spaces. For larger feature sizes, the MEEF is near 1.0 for all resists, controlled by the aerial image MEEF. However, as the feature sizes approach the resolution limit, the characteristic skyrocketing MEEF is observed. (In fact, this dramatic increase in MEEF for smaller features can be used as one definition of resolution: the smallest feature size that keeps the MEEF below some critical value, say 3.) As can be seen from the figure, the major impact of resist contrast is to determine at what feature size the MEEF begins its dramatic rise. Above k_1 values of about 0.6, MEEF values are near 1.0 or below, and resist contrast has little impact. Below this value, the MEEF rises rapidly. Resist contrast affects the steepness of this rise.

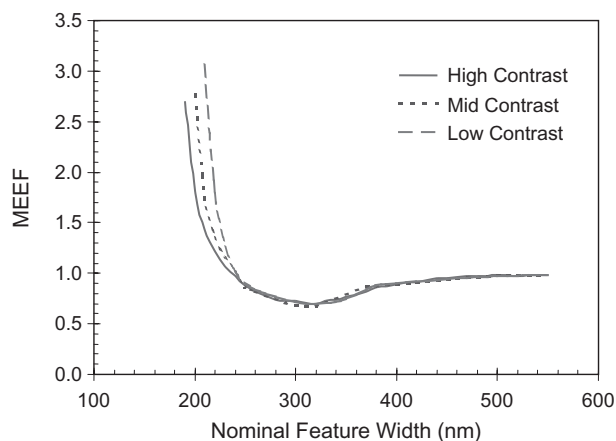


Figure 8.37 Resist contrast affects the mask error enhancement factor (MEEF) dramatically near the resolution limit

Obviously, resist contrast has a huge effect on the MEEF for features pushing the resolution limits. At a k_1 value of 0.5 for the example shown here, the MEEF for the high-contrast resist is 1.5, for the mid-contrast resist it is 2.0, and for the low-contrast resist the MEEF has grown to greater than 3.0. As is seen in so many ways, improvements in resist contrast can dramatically improve the ability to control high-resolution linewidths on the wafer, in this case in the presence of mask errors.

8.8 Line-End Shortening

As lithography pushes to smaller and smaller features, single number metrics such as the critical dimension (CD) of a feature are not adequate. The three-dimensional shapes of the final printed photoresist features can, in fact, affect the performance of the final electrical devices in ways that cannot be described by variations in a single width parameter of those features. In such cases, more information about the shape of a photoresist pattern must be measured in order to characterize its quality. One very simple example is known as *line-end shortening* (LES).

Consider a single, isolated line with width near the resolution limit of a lithographic process. Considerable effort is usually required to develop a process that provides adequate CD control over a range of processing errors (focus, exposure, mask errors, etc.). Although such a feature is generally considered to be one-dimensional (with CD, measured perpendicular to the long line, as the only important dimension), it must, by necessity, have a two-dimensional character at the line end. An important question then arises: for a process where control of the linewidth is adequate, will the shape of the line end also behave acceptably? Often, the answer to this question is *no* due to line-end shortening.

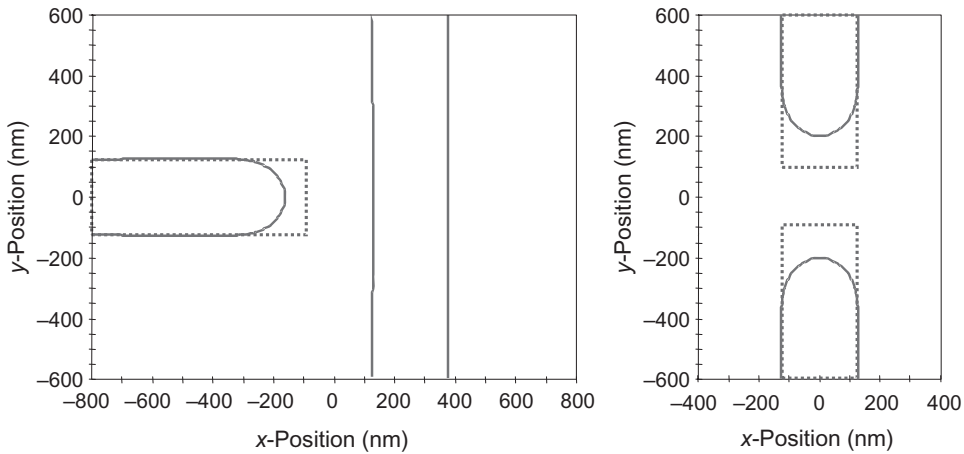


Figure 8.38 Outline of the printed photoresist pattern (solid) superimposed on an outline of the mask (dashed) shows two examples of line-end shortening ($k_1 = 0.6$)

Figure 8.38 illustrates the problem. When the process is adjusted to give the correct CD along the length of the line, the result at the line end will be a pull-back of the resist to produce a foreshortened end. The degree of line-end shortening is a strong function of the line width, with effects becoming noticeable for k less than about 0.8. At first glance, it may seem that a solution to this problem is straightforward. If the printed image of a line end is shorter than the drawn pattern on the mask, simply extending the mask by the amount of line-end shortening would solve the problem. Of course, since the degree of LES is feature size dependent, proper characterization would be required. Most commercial optical proximity correction or design rule checking software today can automatically perform such corrections on the design before the photomask is made (see Chapter 10). This solution ignores two very important problems, however. First, what happens when the end of the line is in proximity to another feature, as in Figure 8.38, and second, how does the degree of LES vary with processing errors? For the first problem, a simple extension of the line may not work. Thus, more complicated corrections (such as increasing the width of the line near the end) are required. For the second problem, a more complete characterization of line-end shortening is needed.

8.8.1 Measuring LES

In order to characterize line-end shortening, the first step is to find a way to measure it. Since LES is fundamentally an error of the resist pattern relative to the design, it cannot be independently measured. Instead, it must be measured as the difference between two measurements, such as a measure of the line-end position relative to another feature. A very simple approach is to use a test structure such as those shown in Figure 8.38 where the line-end shortening is considered to be proportional to the width of the gap between

the end of the line and the edge of the nearby perpendicular line, or between two butting line ends. But this gap alone does not tell the whole story. Changes in the process (such as focus and exposure) will affect the gap width as well as the width of the isolated line.

An interesting approach to normalizing the relationship between linewidth and line-end shortening is to plot the gap width from a structure like that in Figure 8.38 as a function of the resist linewidth over a range of processing conditions. Based on the simple behavior of a pattern of lines and spaces where the resist linewidth plus the spacewidth will always be equal to the pitch, one can establish the ideal, linear imaging result for this case. For the pattern in Figure 8.38, the ideal result should be a straight line with equation $\text{linewidth} + \text{gapwidth} = 500 \text{ nm}$, where the designed linewidth and gap width are both 250 nm. As an example, the data from a focus–exposure matrix are plotted in Figure 8.39 using this technique. Interestingly, all of the data essentially follow a straight line which is offset from the ideal, no line-end shortening result. The vertical offset between the ideal line and a parallel line going through the data can be considered the effective line-end shortening over the range of processing conditions considered. Although not perfect, this result shows that the variables of focus and exposure do not influence the effective LES to first order. The fact that the data form a line which is not exactly parallel to the ideal line simply indicates that, to second order, the LES does not exhibit the same process response to these variables as does the linewidth.

The gap width-versus-linewidth approach to characterizing the effective line-end shortening still ignores the three-dimensional nature of line-end effects. Processing changes, especially focus, can alter the shape as well as the size of a photoresist feature. For the

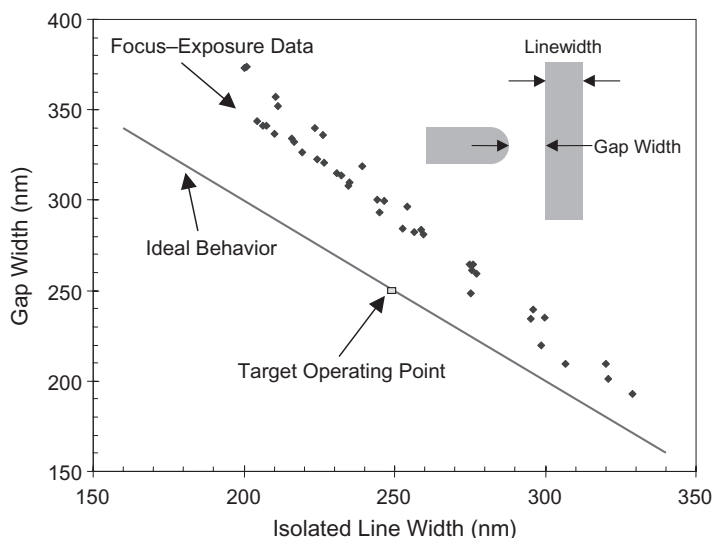


Figure 8.39 Line-end shortening can be characterized by plotting the gap width of a structure like that in the insert as a function of the isolated linewidth under a variety of conditions. As shown here, changes in focus and exposure produce a linear gap width versus linewidth behavior

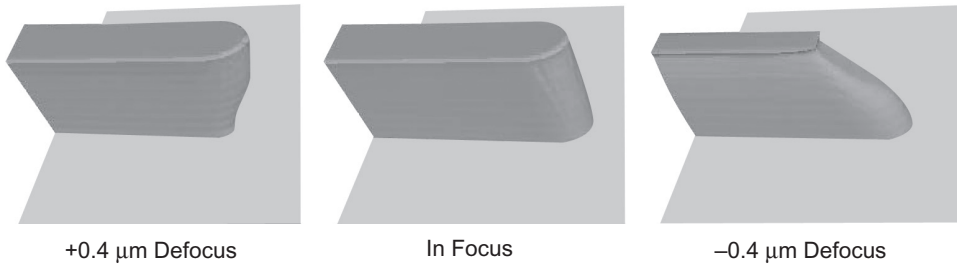


Figure 8.40 Simulated impact of focus on the shape of the end of an isolated line (250-nm line, $NA = 0.6$, $\sigma = 0.5$, $\lambda = 248$, positive focus defined as shifting the focal plane up)

case of a resist line cross section, the sidewall angle of the resist pattern is reduced when out of focus. What is less obvious is that the end of a line is even more sensitive to focus errors than the line itself. Figure 8.40 shows how errors in focus can change the three-dimensional shape of a line end. Obviously, any metrology designed to measure line-end shortening will almost certainly be affected by the shape changes depicted in Figure 8.40.

8.8.2 Characterizing LES Process Effects

Any line whose width is near the resolution limit of a lithographic process will usually exhibit significant line-end shortening. The end of the line is made up of two 90° corners. But as these corners pass through an imaging process of limited resolution, the corners will, by necessity, round. If the feature width is less than the sum of the two corner-rounding radii, the end of the line will pull back due to this corner rounding. There are actually three major causes of this corner rounding, all of which contribute to the final LES magnitude.

The primary corner rounding, and thus line-end shortening, mechanism is the diffraction limitation of aerial image formation. To first order, the aerial image corner-rounding radius is about equal to the radius of the point spread function (about $0.35\lambda/NA$). Thus, significant line-end shortening is expected for feature sizes smaller than about $k_1 = 0.7$. As an example, a 180-nm isolated line imaged at 248 nm with a numerical aperture (NA) of 0.688 (giving a scaled feature size of $k_1 = 0.5$) and with a partial coherence of 0.5 will produce an aerial image with nearly 50 nm of LES. The amount of corner rounding is a strong function of k_1 . Figure 8.41a shows the effect of NA on the line-end shortening of the aerial image of a 180-nm-sized structure like that of Figure 8.38. Obviously, a higher NA (just like a lower wavelength or larger feature size) produces less LES. The influence of partial coherence (Figure 8.41b) is less dramatic but still significant.

LES is also influenced by corner rounding on the reticle. Mask making involves lithography processes that, due to their inherent resolution limits, inevitably produce rounded corners. At the end of a line, rounded corners on the mask reduce light intensity transmitted through the mask in a manner nearly identical to a foreshortening of the mask line. Equating the area lost due to mask corner rounding of radius R (wafer dimensions) with

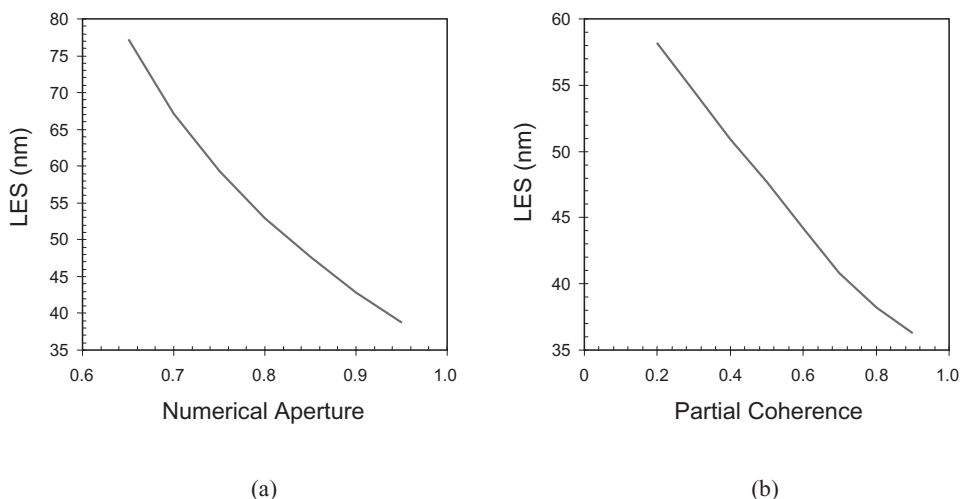


Figure 8.41 Response of line-end shortening (LES) to imaging parameters (130-nm isolated line, $\lambda = 248$ nm): (a) numerical aperture ($\sigma = 0.5$), and (b) partial coherence (NA = 0.85)

an equivalent area due to a foreshortened mask, an approximate LES due to mask corner rounding can be estimated, for a space of width w , as

$$LES_{\text{mask}} \approx \frac{2R^2 \left(1 - \frac{\pi}{4}\right)}{w} \approx \frac{0.43R^2}{w} \quad (8.82)$$

A similar result is expected for lines. Thus, mask corner rounding leads to an effectively shorter mask line, which of course will lead directly to a shorter line printed in resist. As an example, consider a mask-making process with a corner-rounding radius of 20 nm (wafer dimensions). For an 80-nm feature, the amount of LES due to the mask corner rounding can be estimated at 2 nm. If the mask corner rounding is increased to 40 nm, the impact of the mask on LES grows to 9 nm. If the maximum allowed LES due to mask corner rounding is set to be a fixed fraction of the minimum feature size, Equation (8.82) shows that the maximum allowed mask corner rounding must also be a fixed fraction of the minimum feature size.

The resist also affects the final degree of line-end shortening. Interestingly, development contrast seems to have only a small influence on the amount of LES of the final resist patterns. Diffusion during PEB, on the other hand, has a very significant effect. Figure 8.42 shows the simulated result of increasing diffusion length (for an idealized conventional resist and for a chemically amplified resist, both with constant diffusivity) on the line-end shortening of a completely isolated line. The diffusion length must be kept fairly small in order for diffusion to only marginally impact the LES. In both cases, the increase in line-end shortening goes as about the diffusion length squared. The reason for the large effect of diffusion on line-end shortening is the three-dimensional nature of

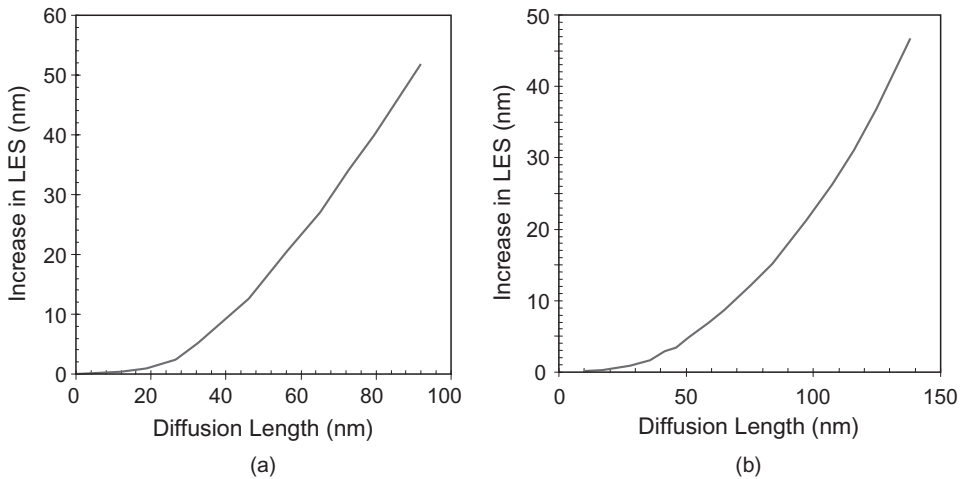


Figure 8.42 Diffusion can have a dramatic effect on line-end shortening (LES) of an isolated line: (a) 180-nm line, $\lambda = 248$ nm, $NA = 0.688$, $\sigma = 0.5$, conventional resist, and (b) 130-nm line, $\lambda = 248$ nm, $NA = 0.85$, $\sigma = 0.5$, chemically amplified resist

the diffusion process: at the end of an isolated line, chemical species from the exposed areas can diffuse from all three sides of the line end (as opposed to one direction for a line edge).

8.9 Critical Shape and Edge Placement Errors

A discussion of CD control assumes that a convenient and accurate dimensional measure of the feature in question is available. For long lines and spaces, and symmetric contact holes, a simple, single metric can be used to describe the difference between the actual result on the wafer and the desired result: the critical dimension error (CDE). The optimum process is that which drives the CDE to zero. But real integrated circuit patterns include numerous complex shapes that defy easy one-dimensional characterization. How does one judge the error in the shape of an arbitrary 2D pattern? For this we must define a *critical shape error* (CSE), the 2D analog to the critical dimension error.

The critical shape error is determined by finding the point-by-point difference between the actual printed resist shape and the desired shape.¹³ For example, Figure 8.43a takes a desired shape (equal to the layout design with an acceptable amount of corner rounding added) and compares it to a top-down image of the printed resist shape. The desired shape is broken down into a collection of measurement points (in this case, equally spaced points are placed around the entire desired shape). Then vectors are drawn from these measurement points to the actual wafer shape, collecting their lengths in a frequency distribution of errors as shown in Figure 8.43b. Once such a distribution is determined, some characterization of the distribution can be used to describe the overall shape error. For example, the average error could be used (CSE_{avg}) or the error which is greater than 90% of the

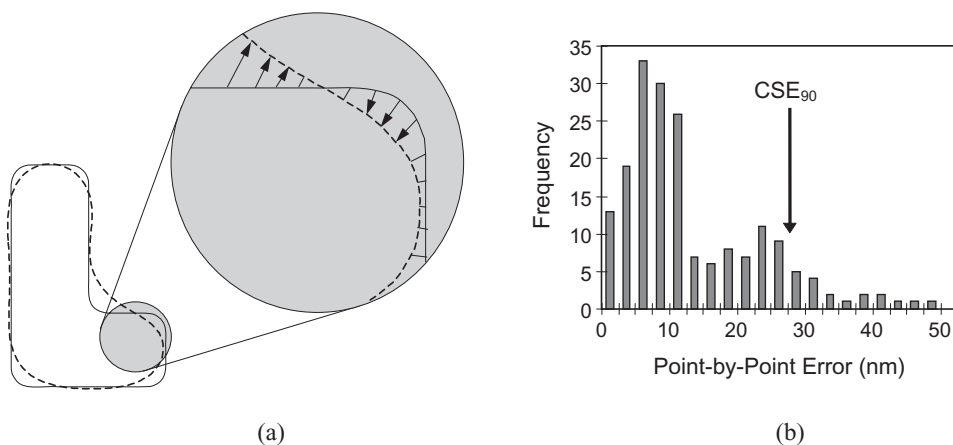


Figure 8.43 Defining a metric of shape error begins with (a) making point-by-point measurements comparing actual (dashed) to desired (solid) shapes, which produces (b) a frequency distribution of errors (unsigned lengths of the vectors are used here). One possible critical shape error (the CSE_{90}) is shown as an example of the analysis of this distribution

point-by-point measurements (CSE_{90}), or some other percentage could also be used. For the distribution in Figure 8.43, some results are given below:

- $CSE_{avg} = 13.5 \text{ nm}$
- $CSE_{80} = 23 \text{ nm}$
- $CSE_{90} = 28 \text{ nm}$
- $CSE_{95} = 32 \text{ nm}$
- $CSE_{99.7} = 46 \text{ nm}$

Several different definitions of the point-by-point error are possible. In the example above, the unsigned length of the vector is used to generate the error histogram. The signed vector could also be used (a mean of zero for the signed vectors would be an indication of a properly sized feature). An alternate approach, useful for features on a ‘Manhattan’ layout, constrains the error vectors to be wholly vertical or horizontal. These Manhattan error vectors are typically called *edge placement errors* (EPE) and are commonly used by optical proximity correction software when characterizing pattern fidelity (see Chapter 10).

8.10 Pattern Collapse

Photoresist profile control has traditionally been defined by making a trapezoidal model of the shape of the resist profile and defining the width, height and sidewall angle of the trapezoid as parameters to be controlled. In general, this approach works well for 1D patterns (long lines and spaces). However, under some circumstances, a different problem can plague the control of profile shape – *pattern collapse*, where seemingly well-formed resist profiles fail mechanically and fall over (Figure 8.44).

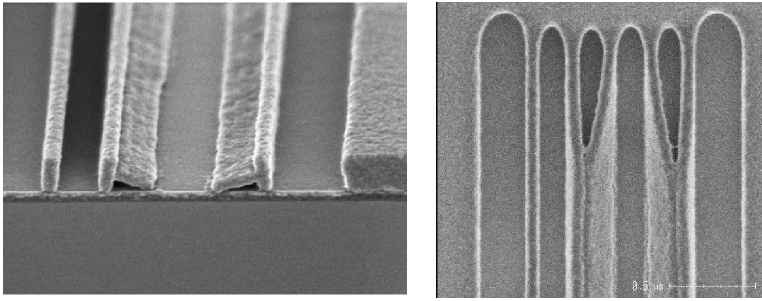


Figure 8.44 Examples of photoresist pattern collapse, both in cross section (left) and top down (Courtesy of Joe Ebihara of Canon)

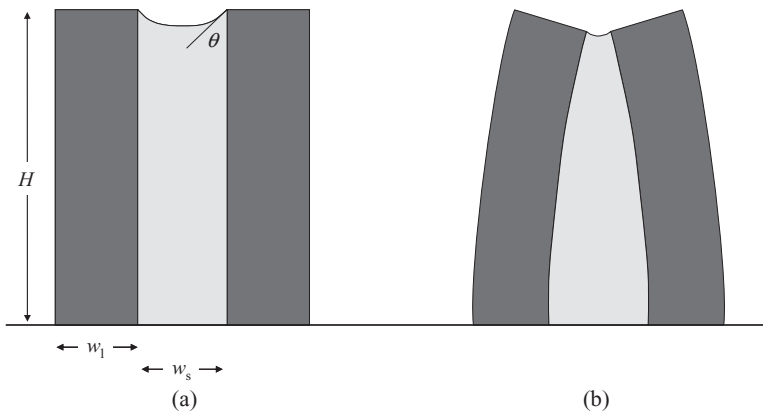


Figure 8.45 Pattern collapse of a pair of isolated lines: (a) drying after rinse leaves water between the lines, and (b) capillary forces caused by the surface tension of water pull the tops of the lines toward each other, leading to collapse

Pattern collapse occurs for tall, narrow resist lines when some force pushing against the top of the line causes the profile to bend and eventually break or peel off the substrate. As one might expect, the taller and narrower the line, the easier it would be to push it over. But where does this pushing force come from? After development, the wafer is rinsed with deionized water and then air-dried. As the water dries, surface tension from the receding water will pull on the top of the resist feature. Of course, if the water is symmetrically receding from both sides of the line, the two forces on each side will cancel. If, however, there is an asymmetry and the water from one side of the line dries off faster than the other side, the resultant force will not be zero.

Figure 8.45a shows what is probably the worst case for generating asymmetric surface tension forces during after-rinse drying: two isolated parallel lines with a small gap between them. The large open areas to each side of the pair quickly dry while the small space between them requires a much longer time for removal of the water. As a result,

there will be a point in the drying cycle where the top meniscus of the water in the gap will line up with the top of the resist features, producing a capillary force, driven by the surface tension of water, that will pull the two resist lines toward each other (Figure 8.45b). If the force is great enough to cause the two resist tops to touch each other, these features are said to have collapsed. Other patterns of lines and spaces (a five-bar pattern, for example) will behave very similarly to this two-bar test pattern for at least the most outside lines.

Tanaka¹⁴ developed a simple cantilever-beam mechanical model for this pattern collapse situation. The capillary pressure (force per unit area) along the side of the resist line covered by water is equal to the surface tension σ of the water–air interface divided by the radius of curvature R of the meniscus. The radius of curvature in turn is determined by the contact angle θ of the water–resist interface and the spacewidth w_s . Before any bending occurs, this curvature will be

$$R = \frac{w_s}{2 \cos \theta} \quad (8.83)$$

For a resist thickness H , the resulting capillary force per unit length of line is

$$F = \frac{\sigma H}{R} = \frac{2\sigma H \cos \theta}{w_s} \quad (8.84)$$

As the resist line starts to bend, however, this force increases. The radius of curvature of the meniscus decreases as the two resist lines come closer together, causing an increase in the capillary force.

Since the force caused by the surface tension is always perpendicular to the air–water interface, a contact angle of zero (very hydrophilic case) causes the maximum force pulling the resist line toward the center of the space. Resists tend to be somewhat hydrophobic, so that contact angles between water and resist are often in the 50–70° range (though these measurements are for unexposed resist, not the partially exposed and deblocked resist along the profile edge where the contact angle should be less). Note that the capillary force increases with the aspect ratio of the space ($A_s = H/w_s$) and with the surface tension of water, which at room temperature is about 0.072 N/m (72 dyne/cm).

The bending of the resist line can be described as an elastic cantilever beam. The important resist material property of mechanical strength is the Young's modulus, E , which is a measure of the stiffness of the resist and is higher for resists with high glass transition temperatures. Resists have been measured to have Young's modulus values in the range of 2–6 GPa with the high end of the range corresponding to novolac resists and the low end of the range for ArF resists.¹⁵ Applying a force F to a line of width w_1 causes that line to move (sway) into the space by an amount δ given by

$$\delta = \frac{3}{2} \left(\frac{F}{E} \right) \left(\frac{H}{w_1} \right)^3 \quad (8.85)$$

Note that the amount of bending is proportional to the cube of the aspect ratio of the resist line, and thus will be very sensitive to this aspect ratio.

As the resist line bends, the pulling of the line by the capillary force is countered by a restoring force caused by the stiffness of the resist. Eventually a 'tipping point' is reached

where the increasing capillary force exceeds the restoring force of the resist and the line collapses. Tanaka calculated this critical point to occur when

$$\frac{E}{\sigma} \leq \frac{4}{w_s} A_l^3 [3A_s \cos \theta + \sin \theta + \sqrt{9A_s^2 \cos^2 \theta + 6A_s \cos \theta \sin \theta}] \quad (8.86)$$

where A_s is the aspect ratio of the space and A_l is the aspect ratio of the line (H/w_l). If the contact angle is less than about 80° and the aspect ratio of the space is high (that is, we are in an interesting regime where pattern collapse is likely to be a problem), the square root can be approximated with the first terms of a Taylor series to give a somewhat more approachable result:

$$\frac{E}{\sigma} \leq \frac{8}{w_s} A_l^3 [3A_s \cos \theta + \sin \theta] \quad (8.87)$$

Equation (8.87) shows what affects pattern collapse and therefore what can be done to try to reduce pattern collapse. Resist chemists could make the resist stiffer by increasing the Young's modulus E (similar to increasing T_g , the glass transition temperature), something that is hard to do without altering other very important resist properties. The water–air surface tension can be reduced by adding surfactant into the rinse liquid. Surfactants can easily reduce the surface tension by a factor of 2 or 3, but may pose contamination problems and can also soften the resist, reducing its Young's modulus. Contact angle of water to the resist also influences collapse. The worst case (maximum value of $3A_s \cos \theta + \sin \theta$) occurs at an angle of $\tan^{-1}(1/3A_s)$ which is generally less than 10° . Thus, hydrophilic resists produce greater pattern collapse. Making the resist more hydrophobic will help (this, incidentally, is also a goal for immersion lithography). The aspect ratio of the space is important, but is less so for hydrophobic resists (small $\cos \theta$). The space width has a direct impact on collapse, with smaller spaces increasing the capillary force and thus the likelihood of collapse. By far the most critical factor is the aspect ratio of the line. Since the tendency to collapse increases as the line aspect ratio cubed, small changes in this factor can have big consequences.

Putting some numbers into the equation, ArF resists will have E/σ in the range of about 35 nm^{-1} assuming no surfactants in the water. Assuming a 60° water–resist contact angle, consider a 100-nm space with a space aspect ratio of 3. The maximum aspect ratio of the line will be 4.3 (a 70-nm linewidth before the line collapses). If the space were shrunk to 50 nm keeping the space aspect ratio at 3, the maximum line aspect ratio drops to 3.4 (a 44-nm linewidth). As another example, when printing a pair of 45-nm equal lines and spaces in a resist of thickness 141 nm, overexposure will cause the patterns to collapse before they reach a 10% CD error. Pattern collapse can become a process window limiter before CD, sidewall angle, or resist loss reach their specifications.

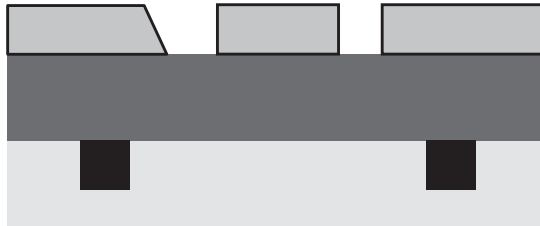
Equation (8.87) can also predict the tendency toward pattern collapse as a function of line/space duty cycle. To simplify, assume that the contact angle is small enough that ignoring the $\sin \theta$ term in Equation (8.87) produces very little error (this approximation is not so bad, even for a 60° contact angle). For this case, a given pitch equal to $w_s + w_l$ will have the minimum tendency to collapse when the linewidth is 50% bigger than the spacewidth (a 3:2 linewidth to spacewidth ratio). All other duty cycles will have an increased chance of pattern collapse. For a fixed duty cycle, the likelihood of collapse

goes as the resist thickness to the fourth power, and as one over the pitch to the third power.

The two-dimensional model used here is really a worst case since it assumes very long lines and spaces. For shorter lines that are connected to another resist feature at one or both ends, the other features can give support to the line, inhibiting the twisting that would be required to make the line collapse. However, it is likely that CMOS device levels like poly and metal 1 will always have some patterns that will behave close to the worst case presented here. And the trend toward smaller feature sizes and constant or even greater aspect ratios will only make the problem of pattern collapse worse in the future.

Problems

- 8.1. A given 90-nm lithography process level exhibits a 0.1-ppm (parts per million) wafer magnification error due to wafer expansion. For a 300-mm wafer, what is the maximum overlay error due to this term if it goes uncorrected? Is the magnitude of this error significant (assume the overlay specifications is 1/3 of the technology node feature size)?
- 8.2. Silicon has a coefficient of linear expansion of about $3 \times 10^{-6}/\text{K}$. Assume that 3 nm of overlay error due to lack of wafer temperature control is acceptable. How well must the temperature of the wafer be controlled for a 200-mm wafer? For a 300-mm wafer?
- 8.3. What are the maximum x - and y -overlay errors due to a 0.3 arc-second reticle rotation error for a 24×28 mm field?
- 8.4. It is possible that the overlay targets, when printed on the wafer, may be asymmetric, as shown below for a frame-in-frame target. What will be the impact of such asymmetry on overlay measurement?



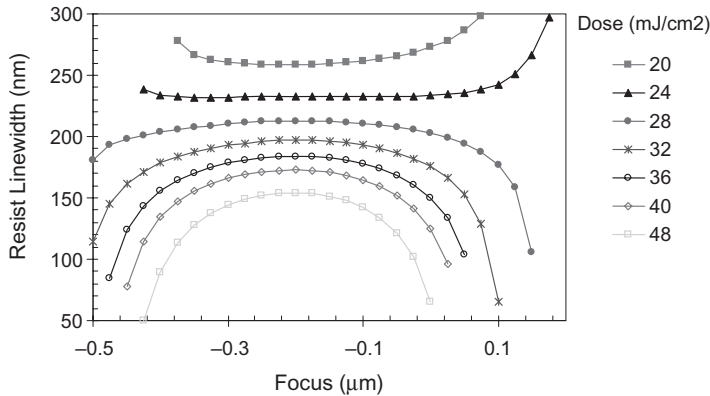
- 8.5. Derive Equations (8.42) and (8.43).
- 8.6. Equations (8.31) and (8.35) both predict CD versus dose, with the later equation a bit more accurate. Plot CD versus dose for these two models on one graph for the case of

$$CD_1 = 80 \text{ nm}, E_1 = 32 \text{ mJ/cm}^2, \text{ and } \left. \frac{d \ln CD}{d \ln E} \right|_{E=E_1} = 0.6$$

At what dose values do the two models disagree with each other in predicted CD by 2%?

8.7. For the Bossung plot given below,

- Approximately where is best focus? Why?
- What is the approximate isofocal bias (numerical value) for a target CD of 180 nm?



- Using Equations (8.50)–(8.52), plot the depth of focus of a 65-nm line as a function of isofocal bias: let $CD_{\text{spec}} = 0.2$, $El_{\text{spec}} = 0.08$, $s_{\text{nom}} = 0.75$, and $\Delta = 0.2 \mu\text{m}$.
- Use the general expression (8.75) to derive the image MEEF for a coherent equal line/space image that uses only the 0th and ± 1 st diffraction orders [that is, derive Equation (8.68)].
- For the two-line pattern collapse problem, what linewidth-to-spacewidth ratio (for a fixed pitch) minimizes the tendency for pattern collapse under the following conditions:
 - Assume that the water–resist contact angle is small enough that the $\sin\theta$ term in Equation (8.87) can be ignored.
 - Assume that the water–resist contact angle is large enough that the $\cos\theta$ term in Equation (8.87) can be ignored.

References

- Sturtevant, J., Allgair, J., Fu, C., Green, K., Hershey, R., Kling, M., Litt, L., Lucas, K., Roman, B., Seligman, G. and Schippers, M., 1999, Characterization of CD control for sub- $0.18 \mu\text{m}$ lithographic patterning, *Proceedings of SPIE: Optical Microlithography XII*, **3679**, 220–227.
- Mack, C.A., 1992, Understanding focus effects in submicron optical lithography, part 3: methods for depth-of-focus improvement, *Proceedings of SPIE: Optical/Laser Microlithography V*, **1674**, 272–284.
- Sethi, S., Barrick, M., Massey, J., Froelich, C., Weilemann, M. and Garza, F., 1995, Lithography strategy for printing $0.35 \mu\text{m}$ devices, *Proceedings of SPIE: Optical/Laser Microlithography VIII*, **2440**, 619–632.
- SEMI Standard SEMI P35-1106, 2006, *Terminology for Microlithography Metrology*, Semiconductor Equipment and Materials International, San Jose, CA.

- 5 Levinson, H.J., 1999, *Lithography Process Control*, SPIE Press Vol. TT28, Bellingham, CA.
- 6 Ausschnitt, C., 1988, Chapter 8: electrical measurements for characterizing lithography, in *VLSI Electronics: Microstructure Science, Vol. 16*, Academic Press, p. 330.
- 7 DeMoor, S., Brown, J., Robinson, J.C., Chang, S. and Tan, C., 2004, Scanner overlay mix and match matrix generation; capturing all sources of variation, *Proceedings of SPIE: Metrology, Inspection, and Process Control for Microlithography XVIII*, **5375**, 66–77.
- 8 Bossung, J.W., 1977, Projection printing characterization, *Proceedings of SPIE: Developments in Semiconductor Microlithography II*, **100**, 80–84.
- 9 Ausschnitt, C.P., Rapid optimization of the lithographic process window, *Proceedings of SPIE: Optical/Laser Microlithography II*, **1088**, 115–133.
- 10 Brunner, T., Martin, A., Martino, R., Ausschnitt, C., Newman, T. and Hibbs, M., 1994, Quantitative stepper metrology using the focus monitor test mask, *Proceedings of SPIE: Optical/Laser Microlithography VII*, **2197**, 541–549.
- 11 Maurer, W., 1996, Mask specifications for 193-nm lithography, *Proceedings of SPIE*, **2884**, 562–571.
- 12 The derivation was first suggested to me by Mark D. Smith.
- 13 Tsudaka, K., *et al.*, 1995, Practical optical mask proximity effect correction adopting process latitude consideration, *MicroProcess '95 Digest of Papers*, 140–141.
- 14 Tanaka, T., Morigami, M. and Atoda, N., 1993, Mechanism of resist pattern collapse during development process, *Japanese Journal of Applied Physics*, **32**, 6059–6064.
- 15 Simons, J., Goldfarb, D., Angelopoulos, M., Messick, S., Moreau, W., Robinson, C., de Pablo, J. and Nealey, P., 2001, Image collapse issues in photoresist, *Proceedings of SPIE: Advances in Resist Technology and Processing XVIII*, **4345**, 19–29.